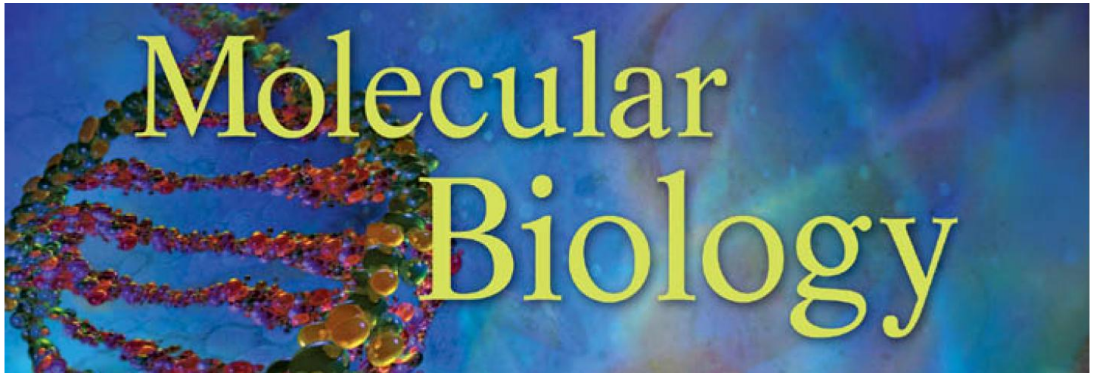


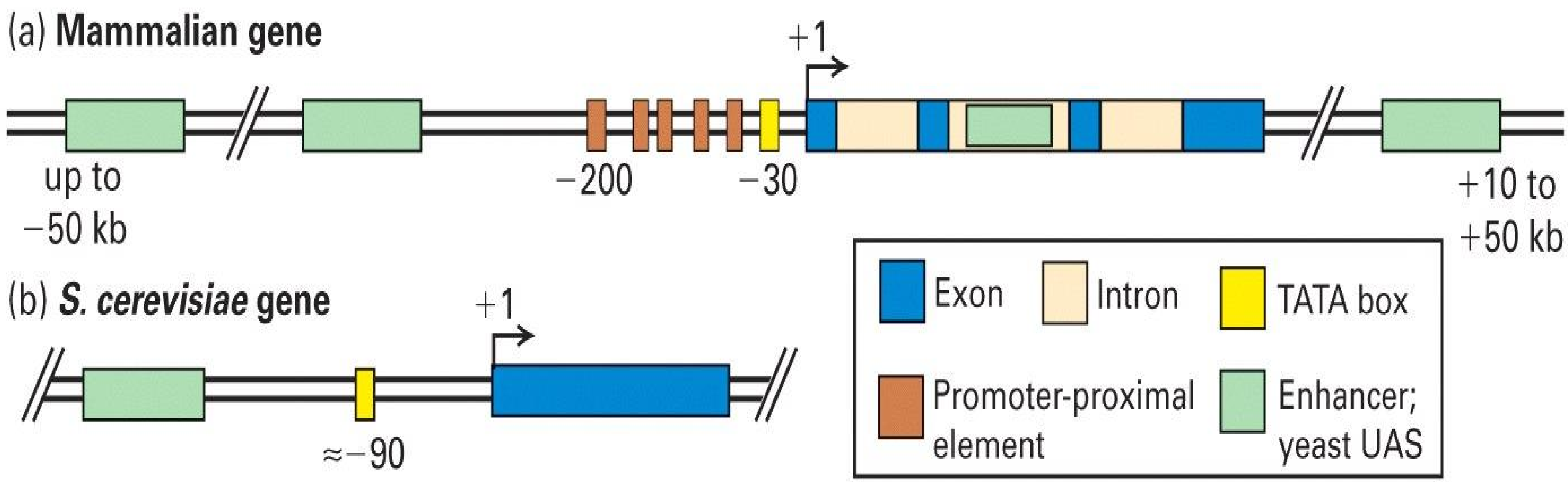
Transcription control in eukaryotes

周明

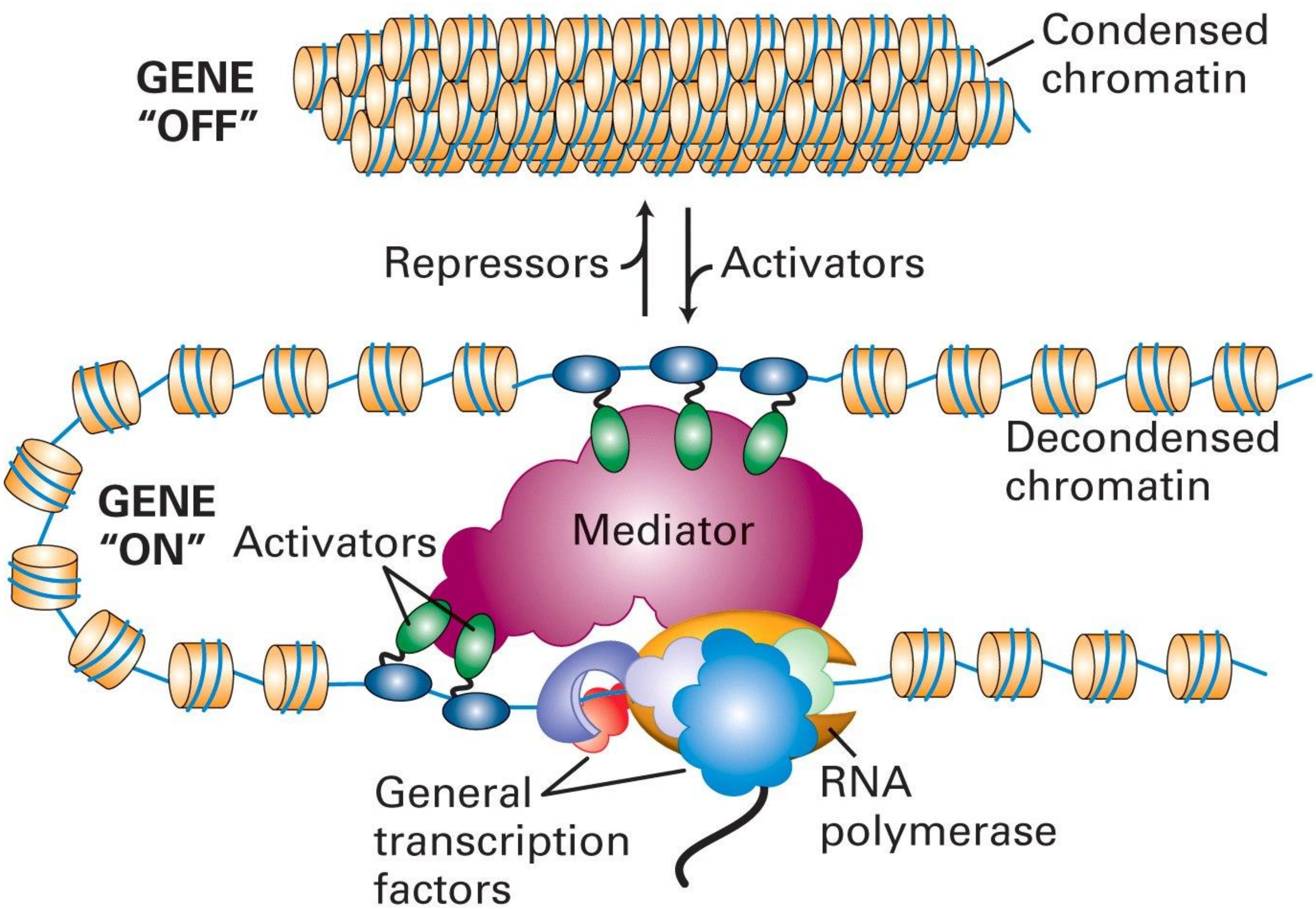
浙江大学生科院 植物所

mingzhou@zju.edu.cn

Transcription control elements in eukaryotes



A schematic picture of transcriptional initiation in eukaryotes



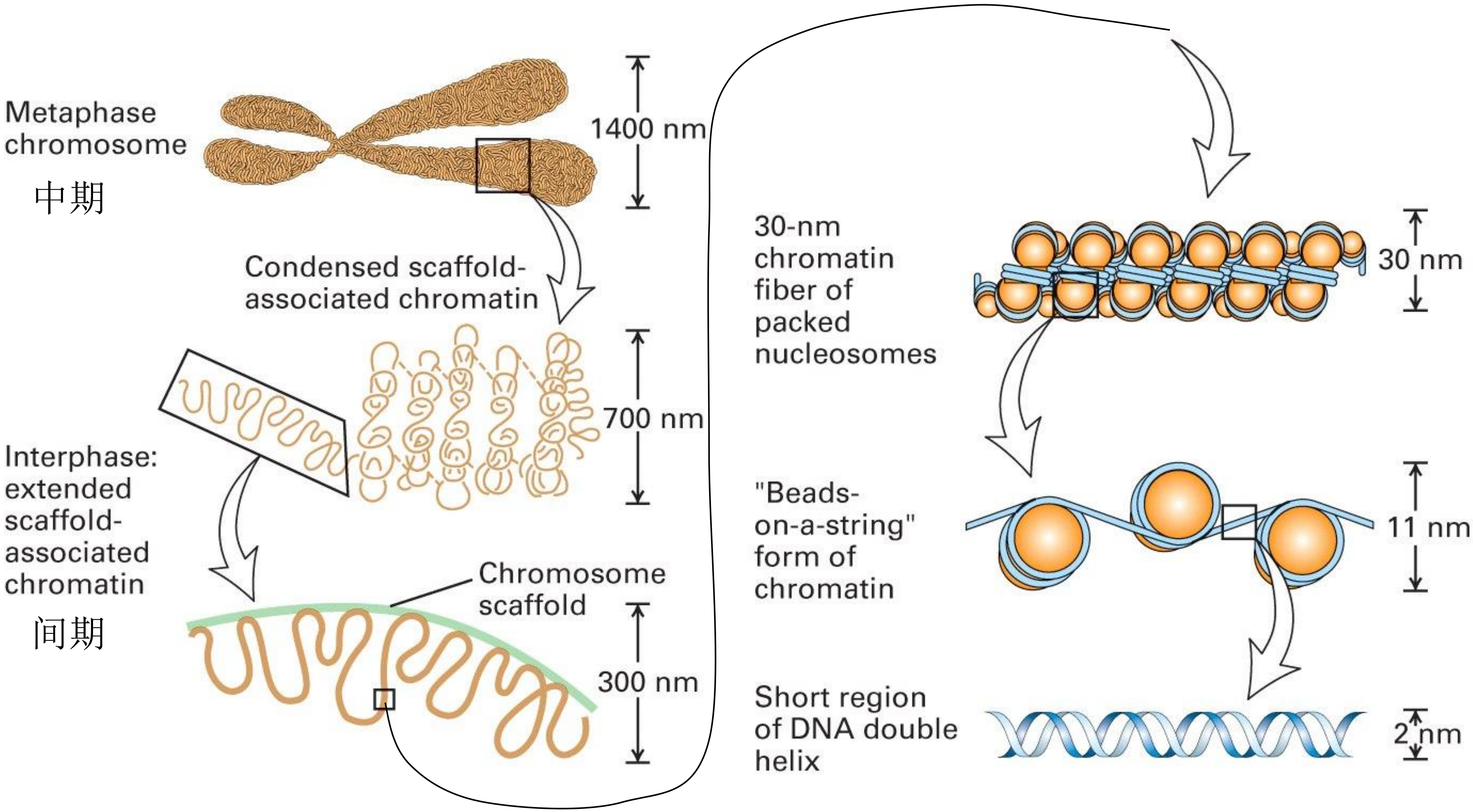
Molecular mechanisms of transcription activation and repression

- 1. Chromatin mediated transcription control
- 2. Transcription control through the Mediator
- 3. Transcription control through DNA methylation
- 4. Regulation of transcription factor activity
- 5. Nuclear receptor

Molecular mechanisms of transcription activation and repression

- 1. Chromatin mediated transcription control
- 2. Transcription control through the Mediator
- 3. Transcription control through DNA methylation
- 4. Regulation of transcription factor activity
- 5. Nuclear receptor

Chromatin structure

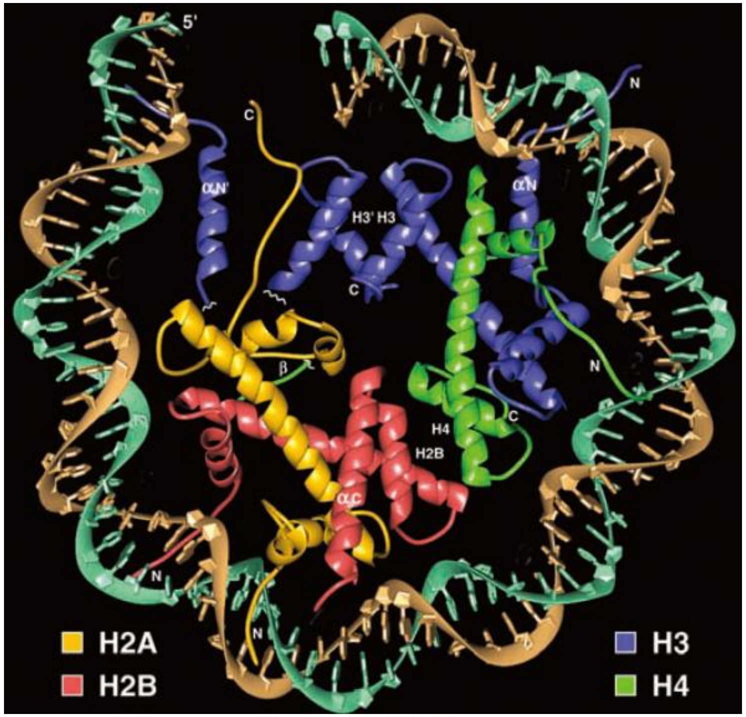


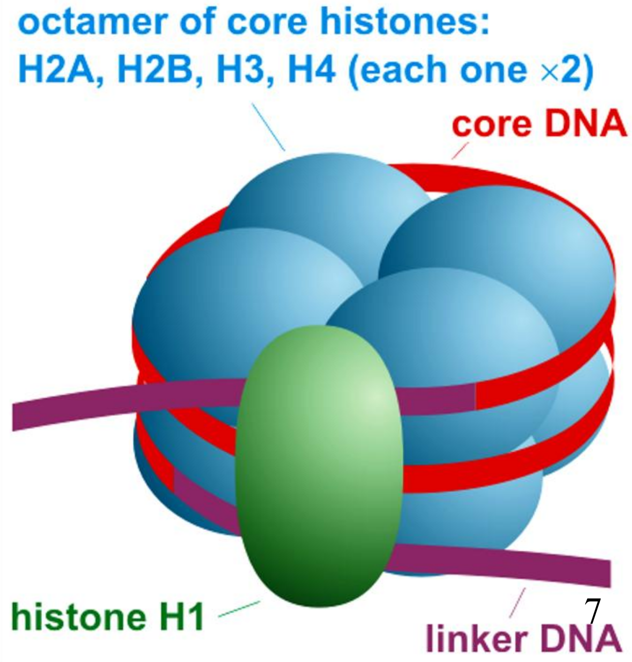
Nucleosome structure

Table 13.1 General Properties of the Histones

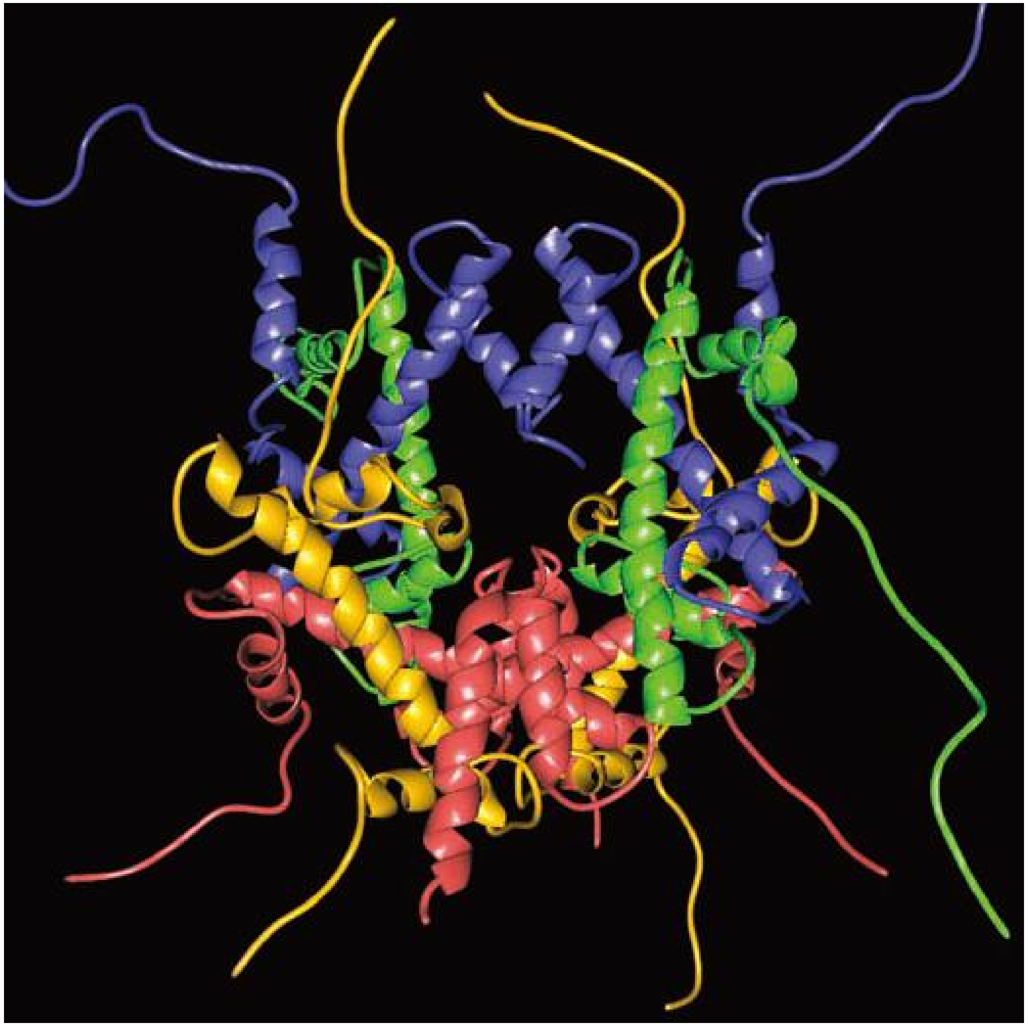
Histone Type	Histone	Molecular Mass (M_r)
Core histones	H3	15,400
	H4	11,340
	H2A	14,000
	H2B	13,770
Linker histones	H1	21,500
	H1°	~21,500
	H5	21,500

Source: Adapted from Critical Reviews in Biochemistry and Molecular Biology, by Butler, P.J.C., 1983. Taylor & Francis Group. LLC., http://www.taylorandfrancis.com

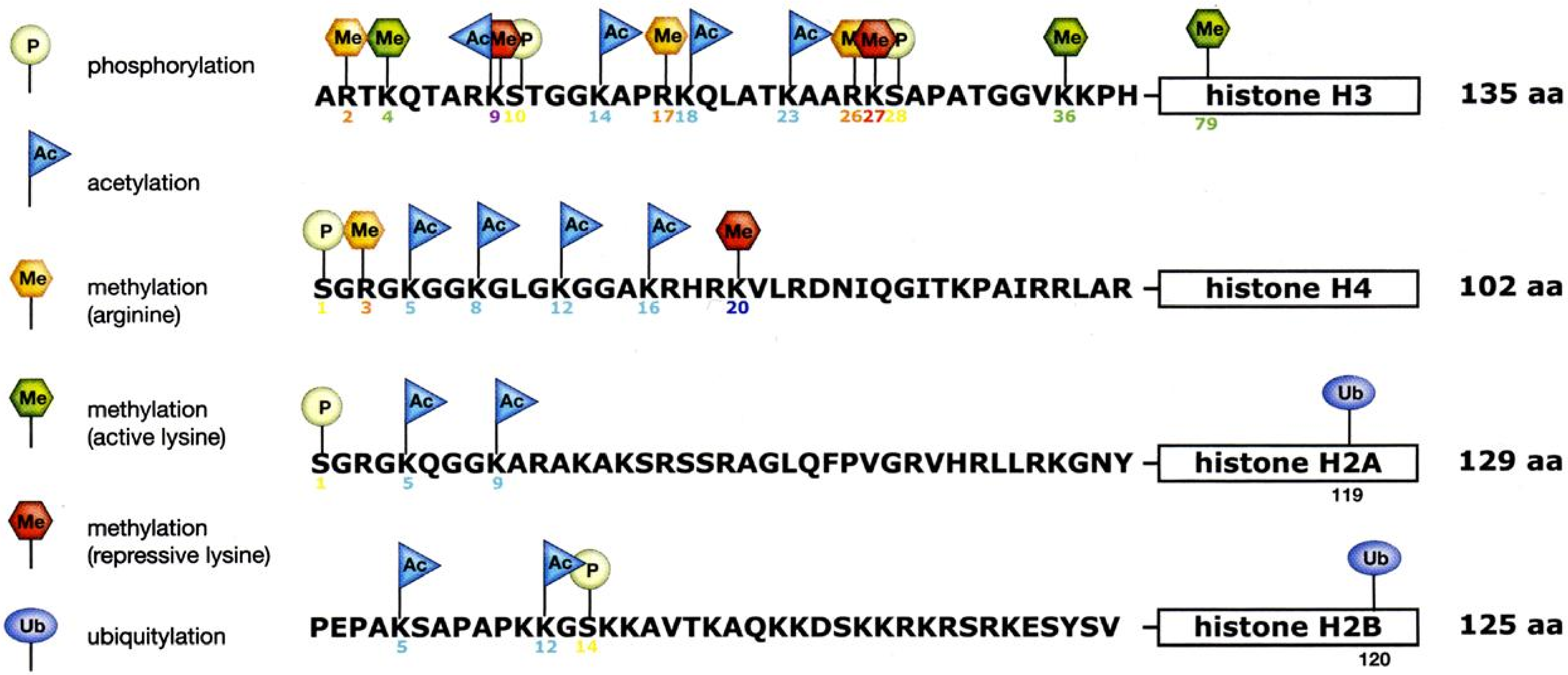




Histone modifications



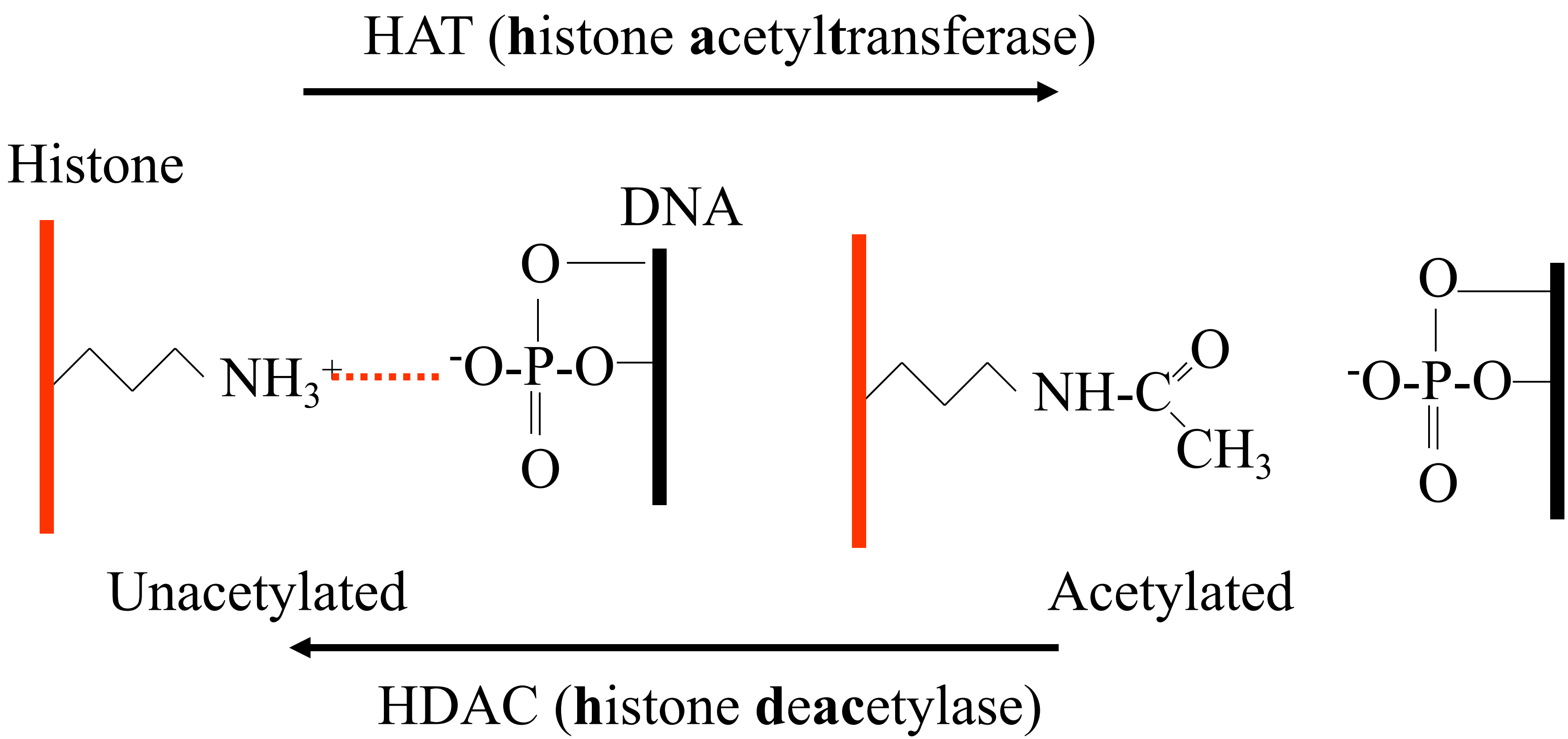
Histone modifications



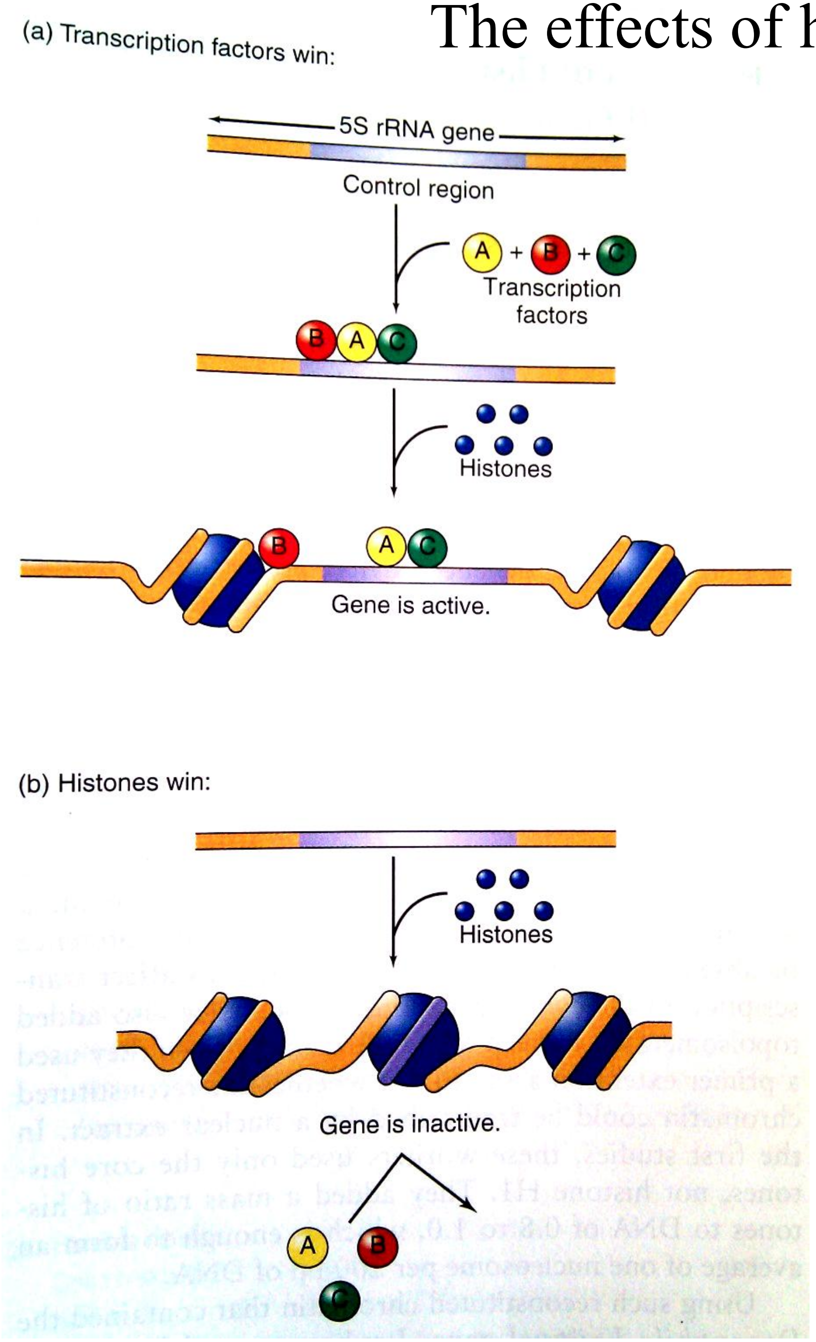
There are over 60 different residues on histones where modifications have been detected.

Acetylation of the N-terminal sequence of histone H3

- ARTKQTARKSTGGKAPRKQL



The effects of histone on gene expression



The race between transcription factors and histones for the 5s rRNA control region

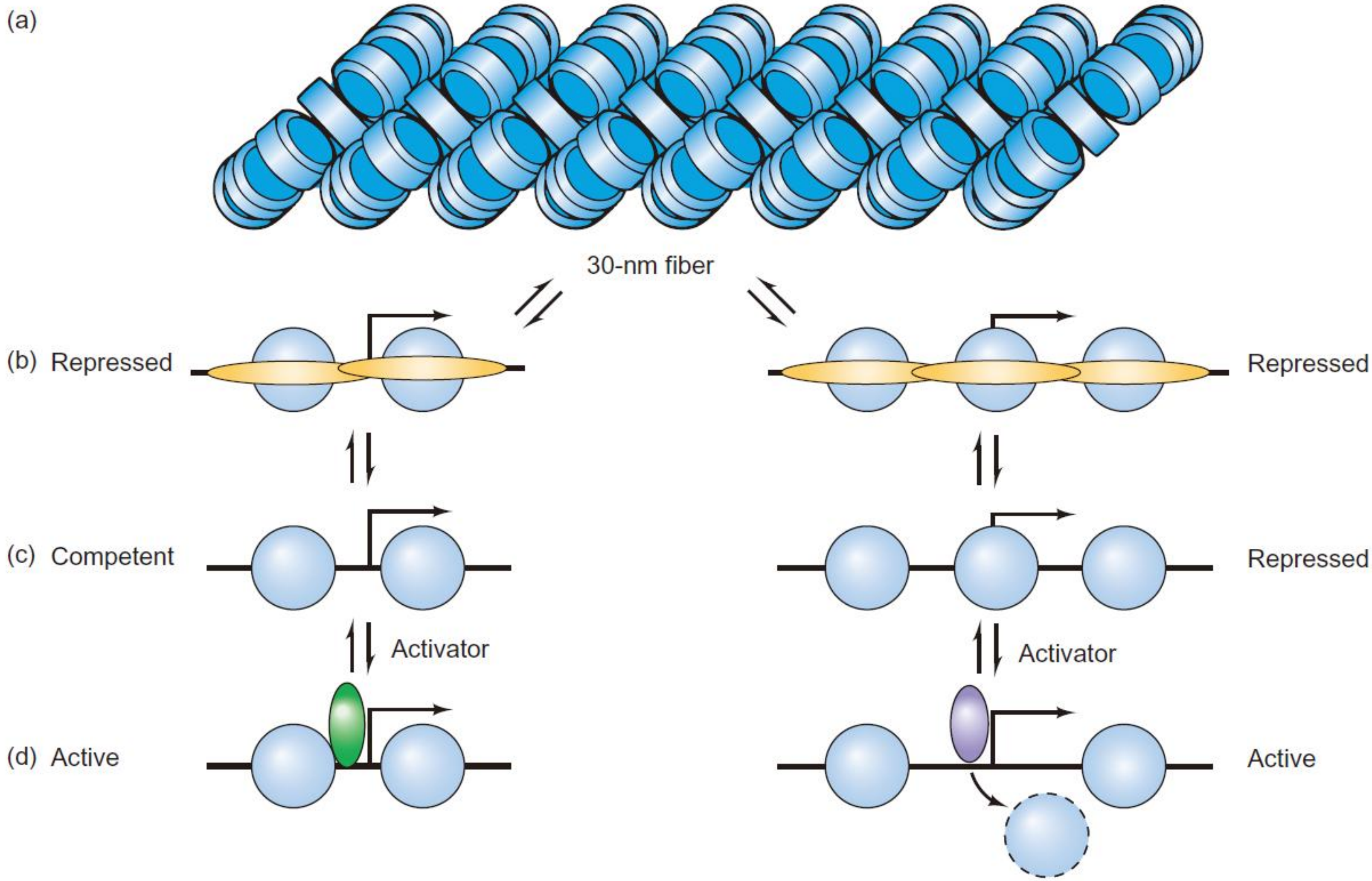
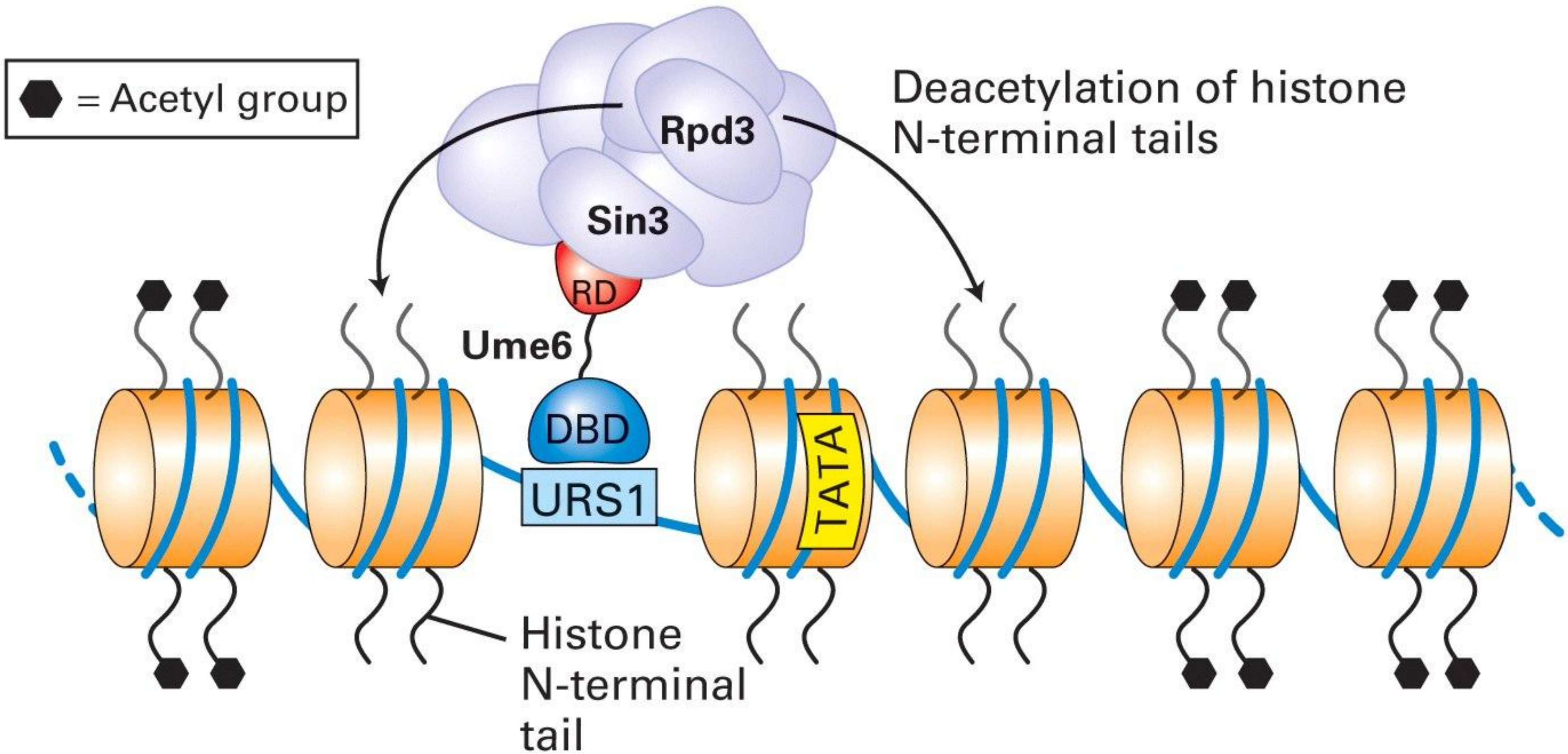


Figure 13.14 A model of transcriptional activation.

(a) Repressor-directed histone deacetylation

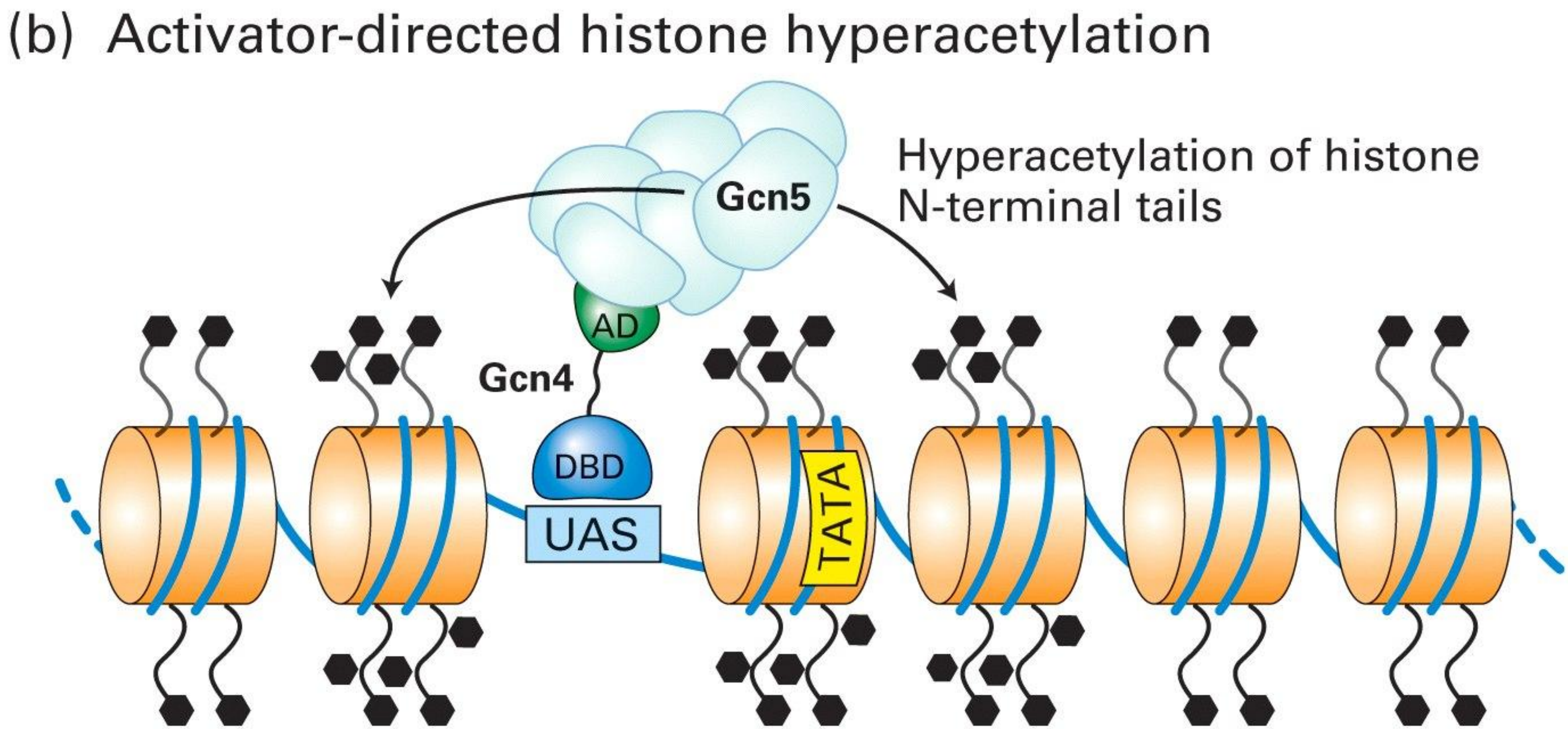


URS1 – Upstream regulatory sequence

DBD and RD – DNA binding and repressor domains of UME6 repressor

RPD3 – yeast histone deacetylase (component of deacetylation complex)

Sin3 – RD binding component of deacetylation complex



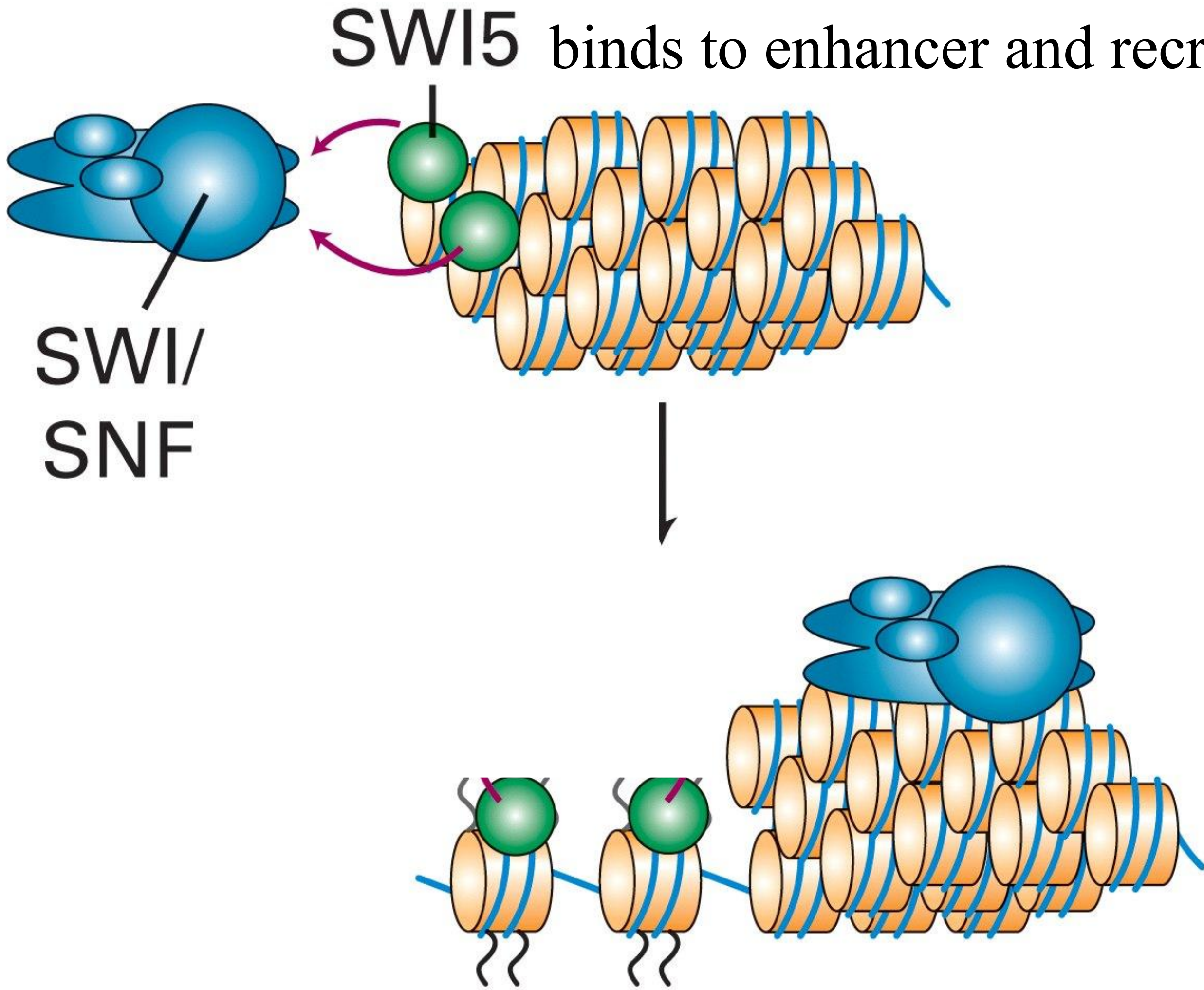
UAS – upstream activation sequence

AD –activation domain of Gcn4 transcription activator

Gcn5 – histone acetylase subunit of acetylation complex

Chromatin remodelling factors

- Chromatin remodellig factors are multiprotein complexes with some subunits showing helicase activity
- Chromatin remodelling complex SWI/SNF transiently dissociates DNA from the surface of nucleosomes, decondensing the chromatin and making the DNA more accessible to transcription factors
- The activity of complex may also result in transcription repression, probably by exposing the histone tails to deacetylases or by assisting in folding of chromatin into higher-order structures

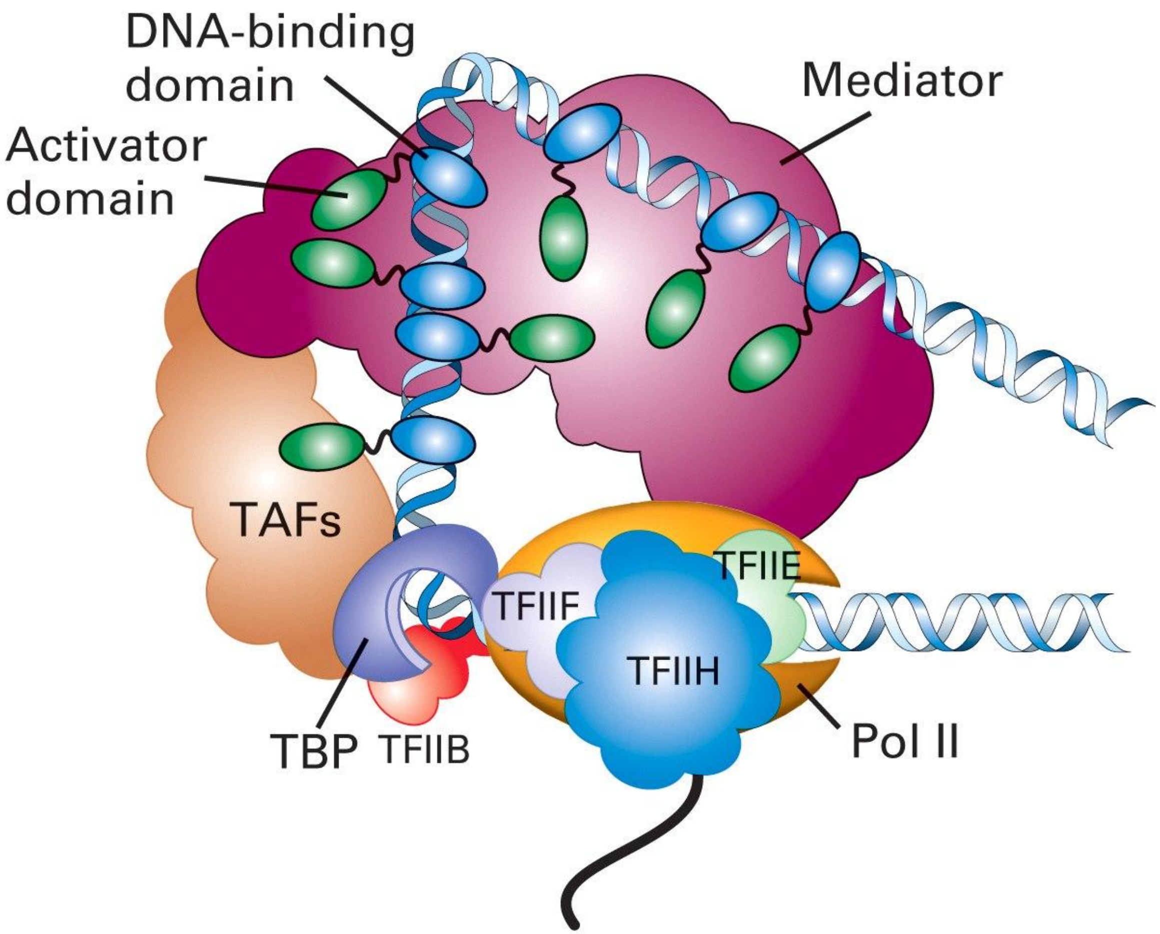


binds to enhancer and recruits SWI/SNF

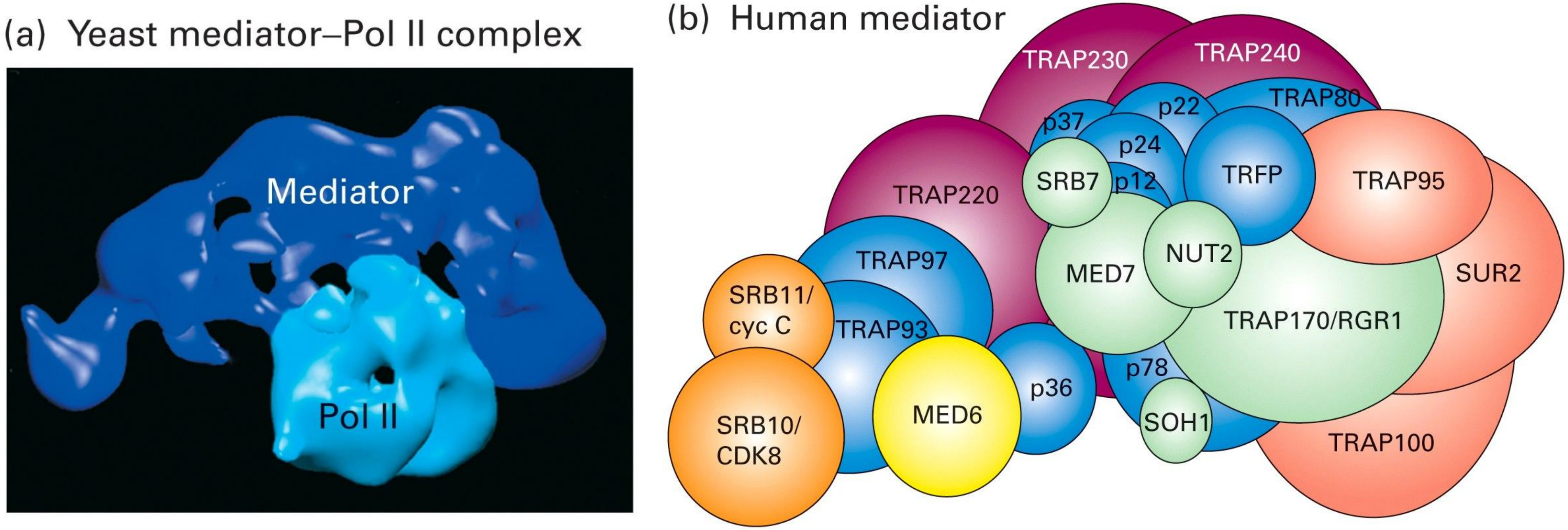
Molecular mechanisms of transcription activation and repression

- 1. Chromatin mediated transcription control
- 2. Transcription control through the Mediator
- 3. Transcription control through DNA methylation
- 4. Regulation of transcription factor activity
- 5. Nuclear receptor

Transcription regulation through Mediator



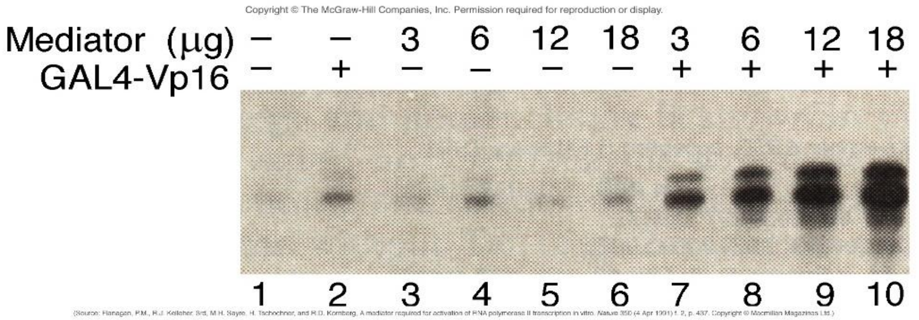
Structure of yeast and human mediator complexes

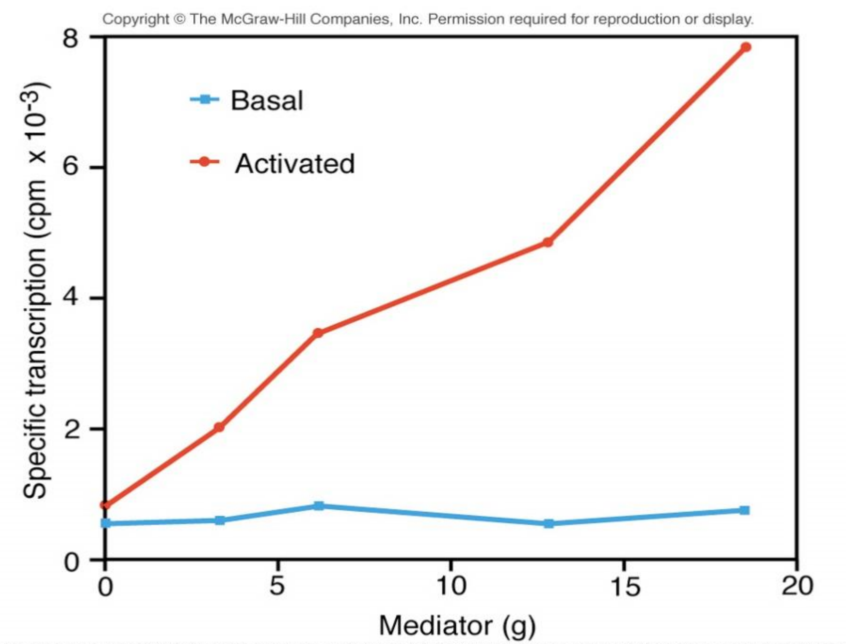


- Composed of ~20 subunits which are arranged in modules
- Some subunits interact with RNA Pol II, others – with activators
- One subunit has histone acetylase activity which might keep the promoter region in hyperacetylated state
- Some subunits are required for expression of all genes ("core subunits") whereas others are required for specific subsets of genes

Proof of mediator

Yeast CYC1 promoter



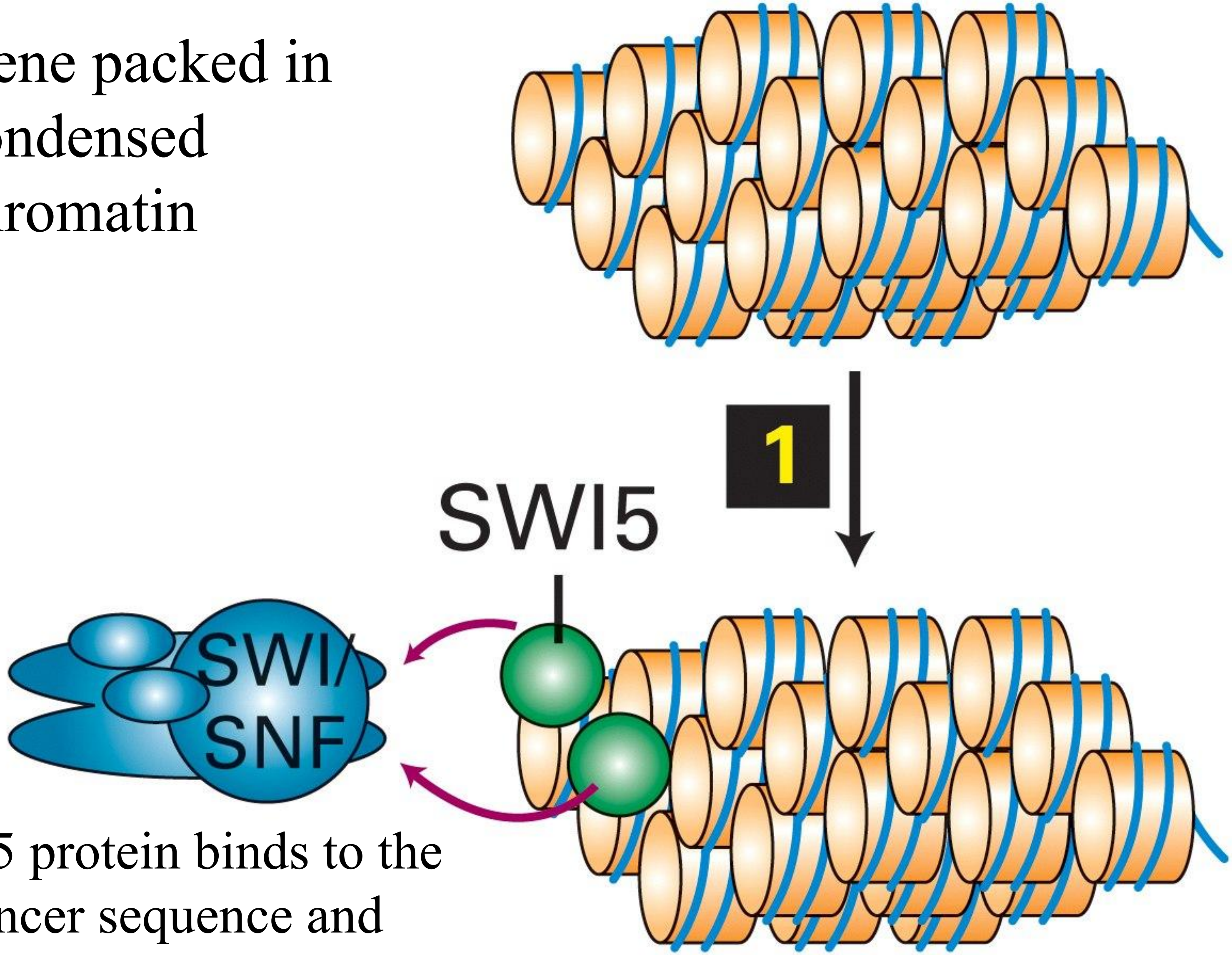


Mediator has no effect on transcription without activator (lane 3-6)
But it greatly stimulated transcription in the presence of the activator
(lane 7-10)

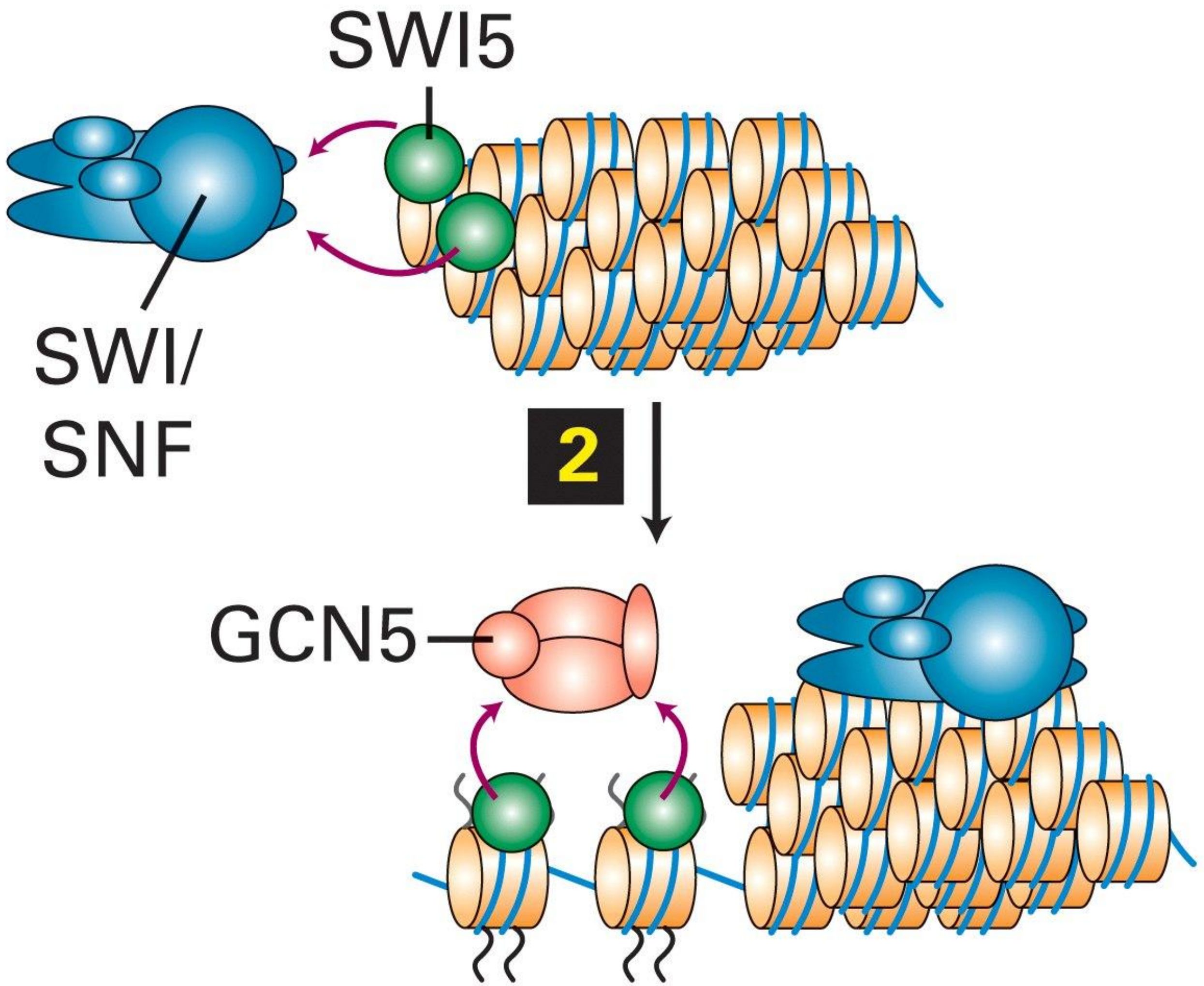
There is a long way from condensed chromatin to mRNA expression...

Example of yeast HO gene activation. HO encodes a site specific nuclease, which initiates mating-type(接合型) switching in haploid yeast cells

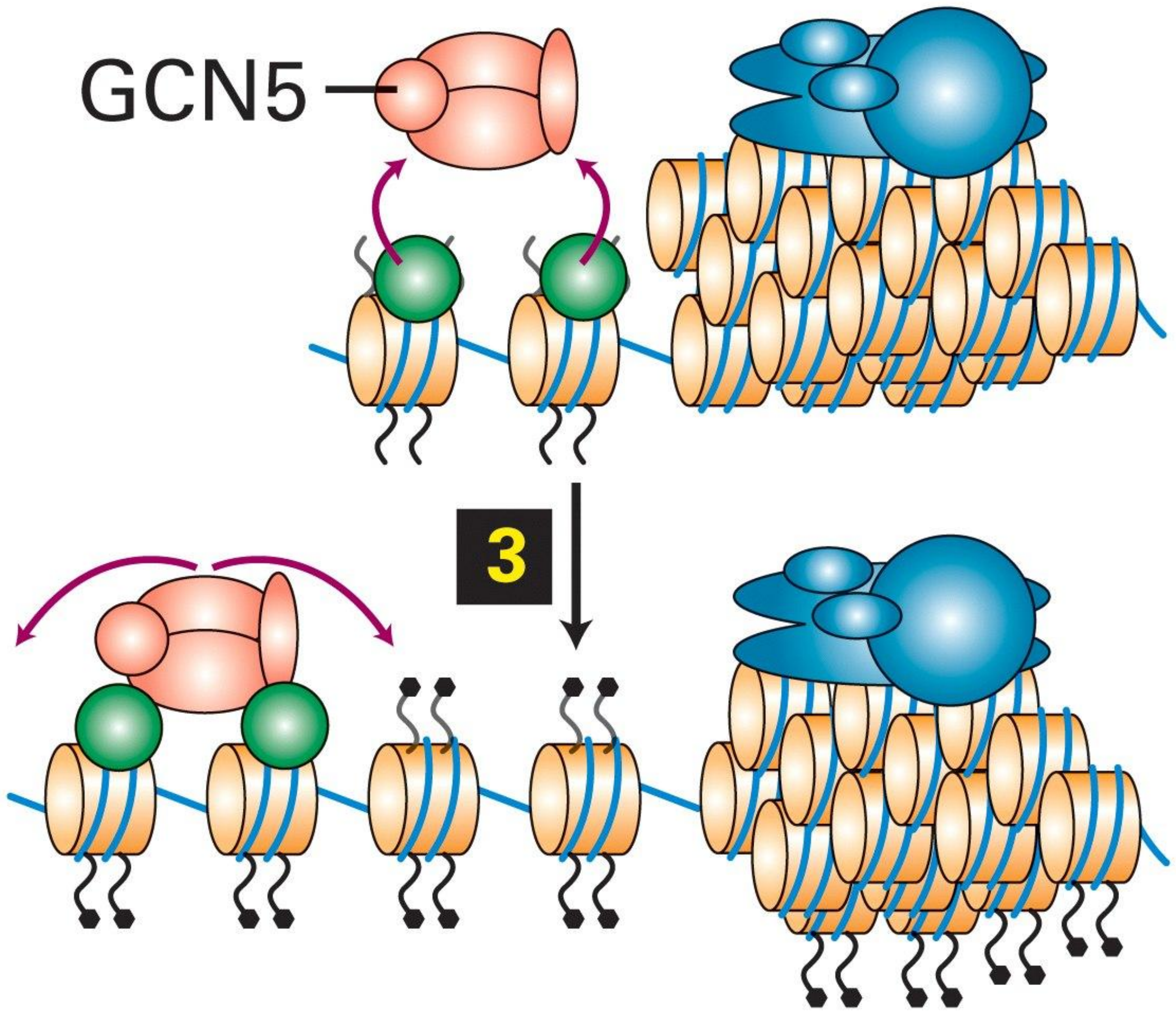
Gene packed in condensed chromatin



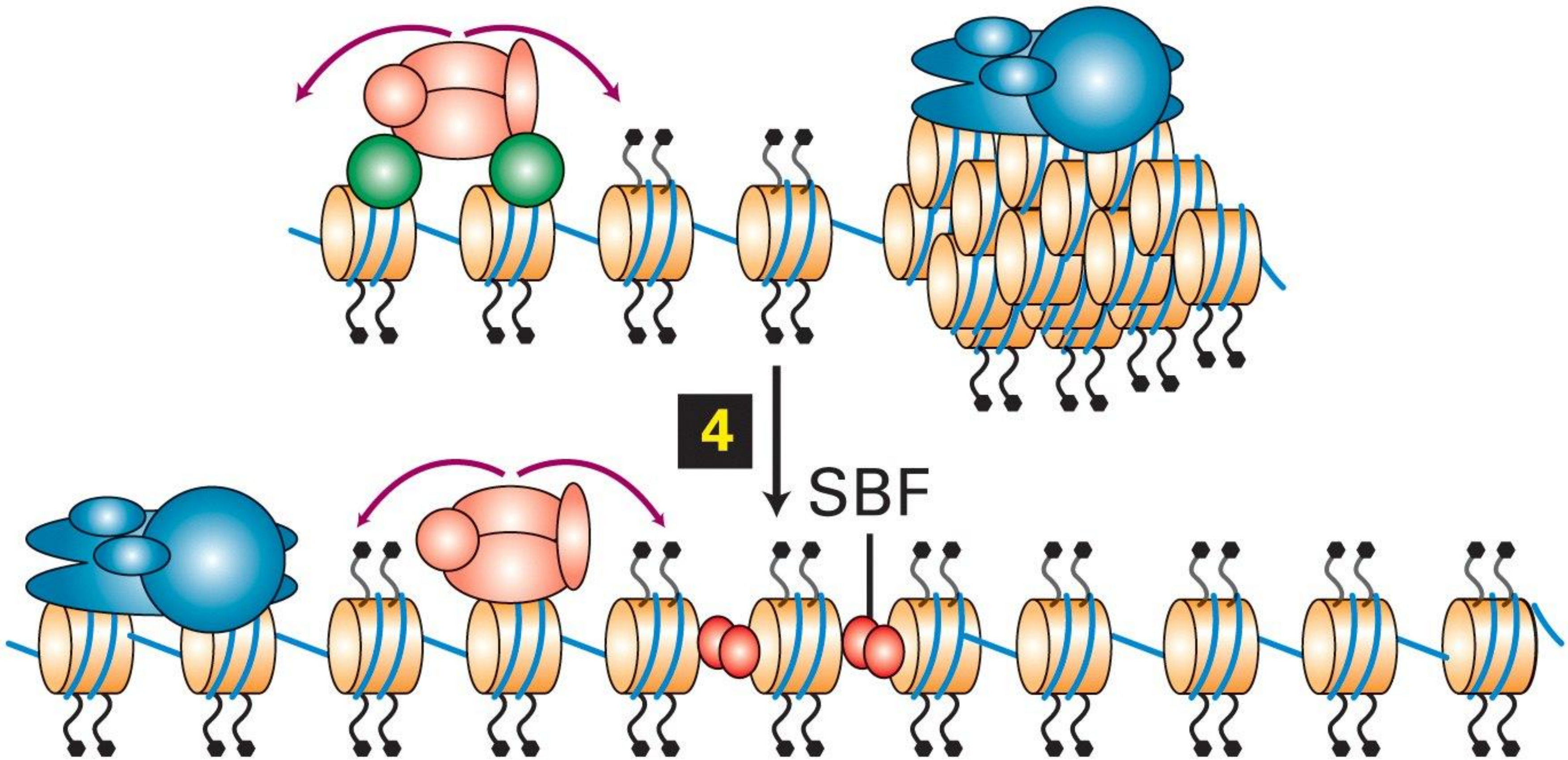
SWI5 protein binds to the enhancer sequence and recruits chromatin remodelling complex



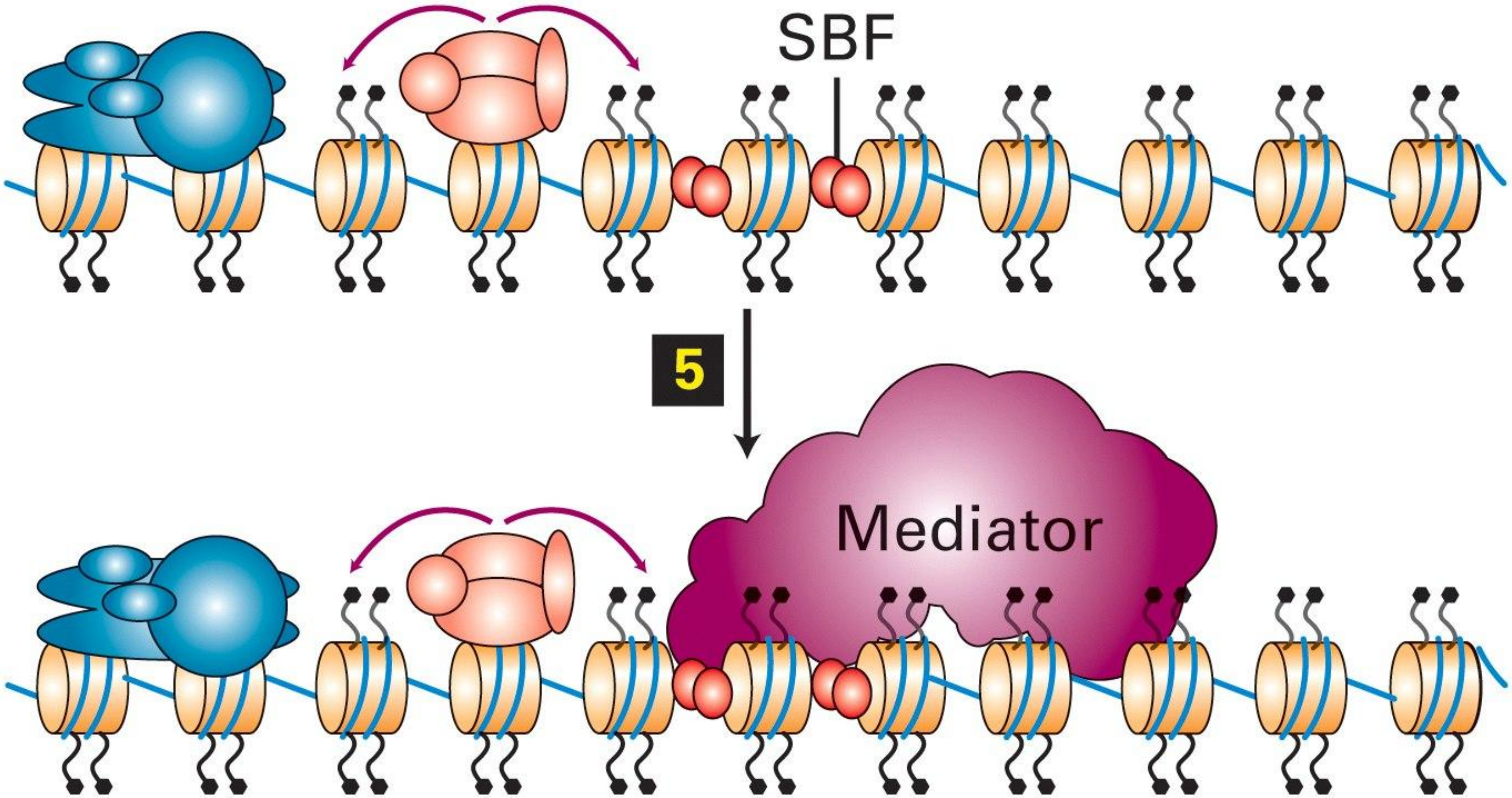
SWI/SNF decondenses chromatin and exposes histone tails.
Histone acetylases get recruited by SWI5



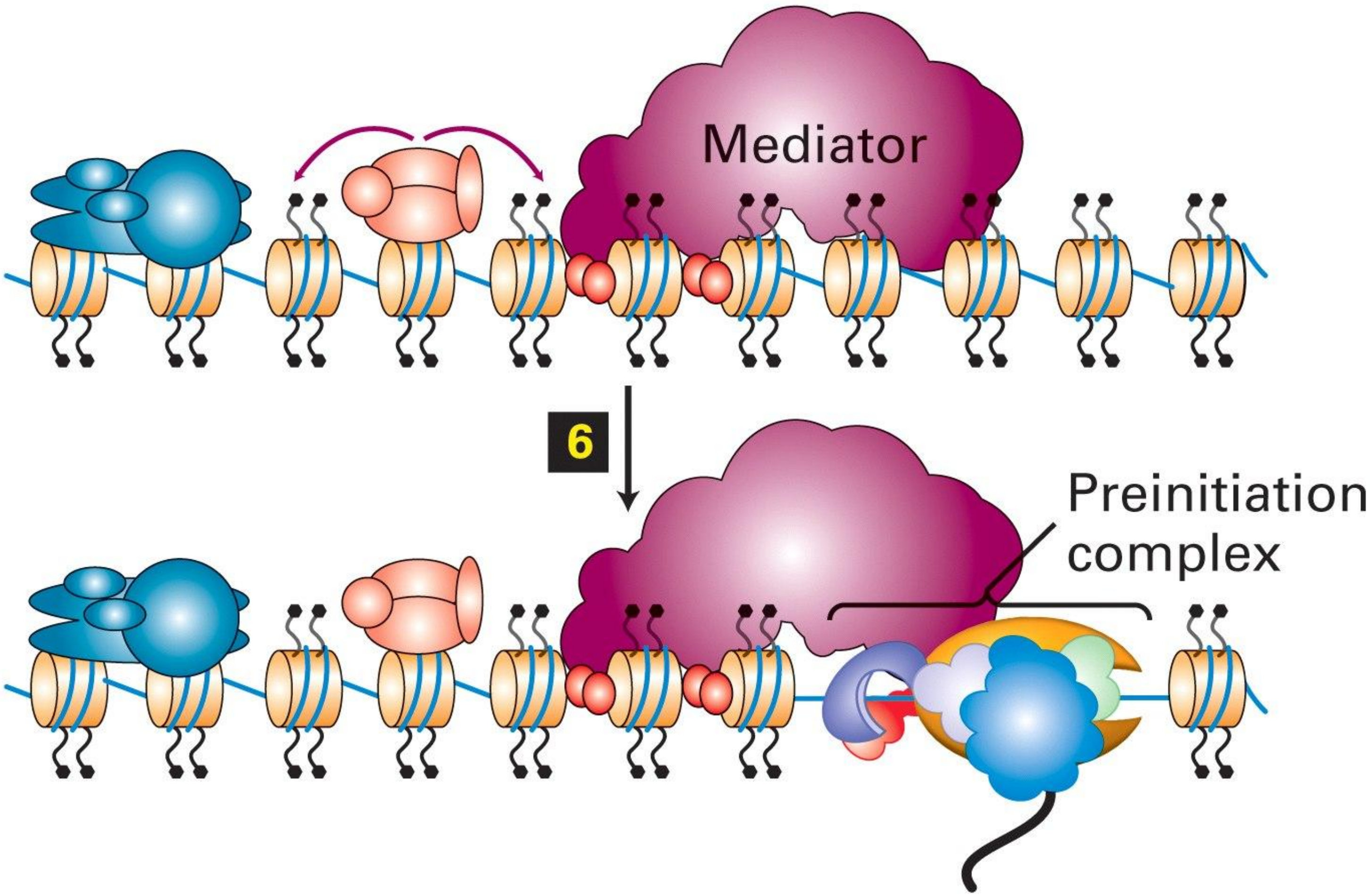
Histone tails get acetylated by GCN5



SBF activator gets bound to promoter proximal elements



Mediator binds to SBF



GTFs and pol II bind to TATA box and form the preinitiation complex

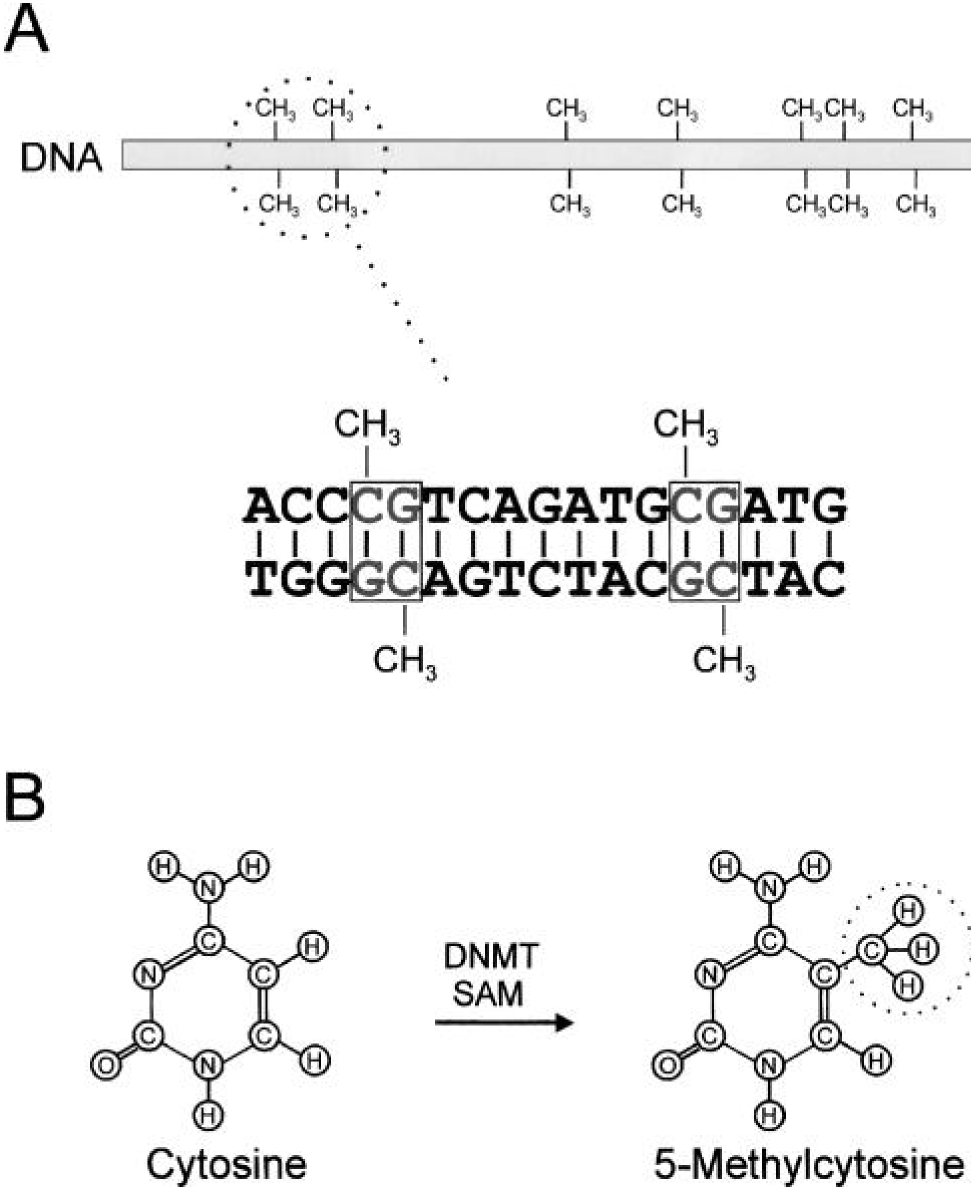
Molecular mechanisms of transcription activation and repression

- 1. Chromatin mediated transcription control
- 2. Transcription control through the Mediator
- 3. Transcription control through DNA methylation
- 4. Regulation of transcription factor activity
- 5. Nuclear receptor

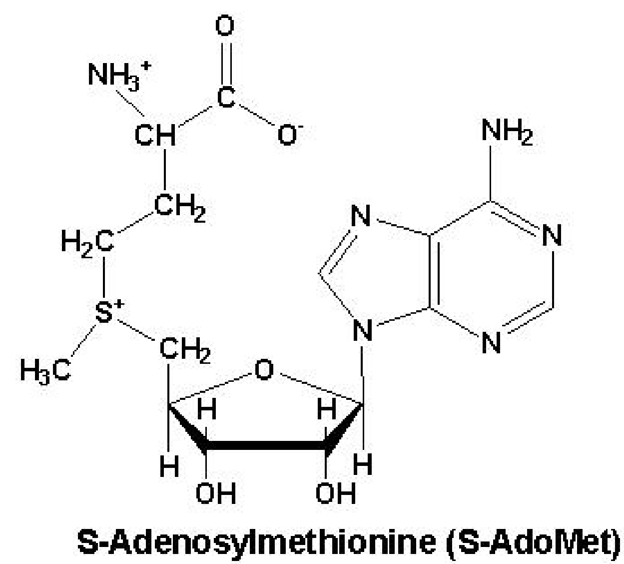
Epigenetic control mechanisms

Methylation of DNA and histones

DNA methylation at CpG islands

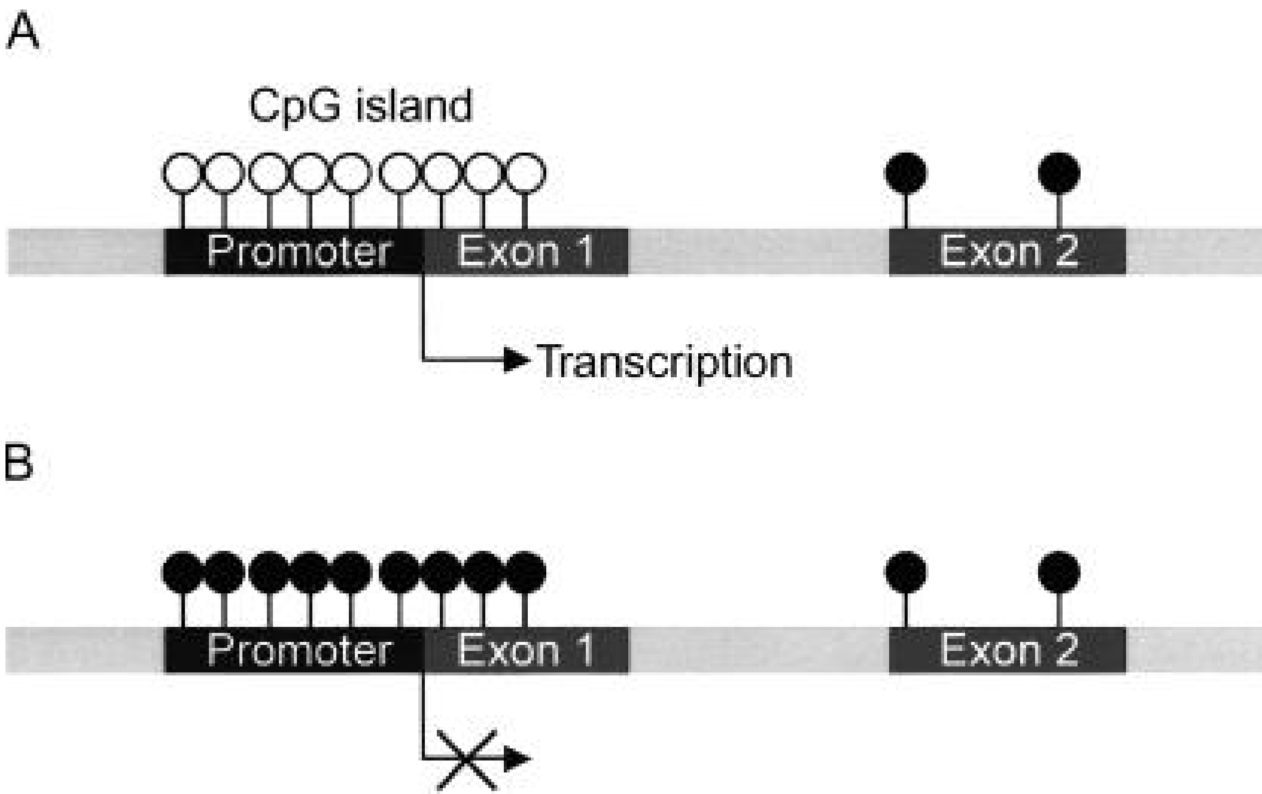


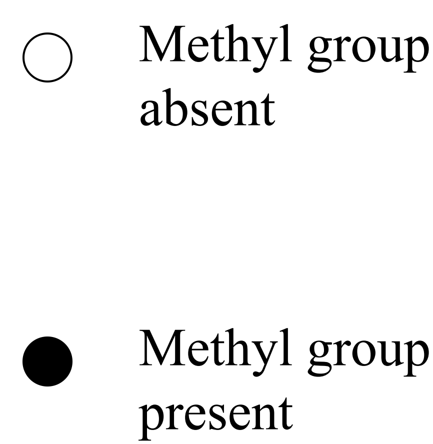
- DNMT – DNA methyltransferase
- SAM: S-adenosylmethionine



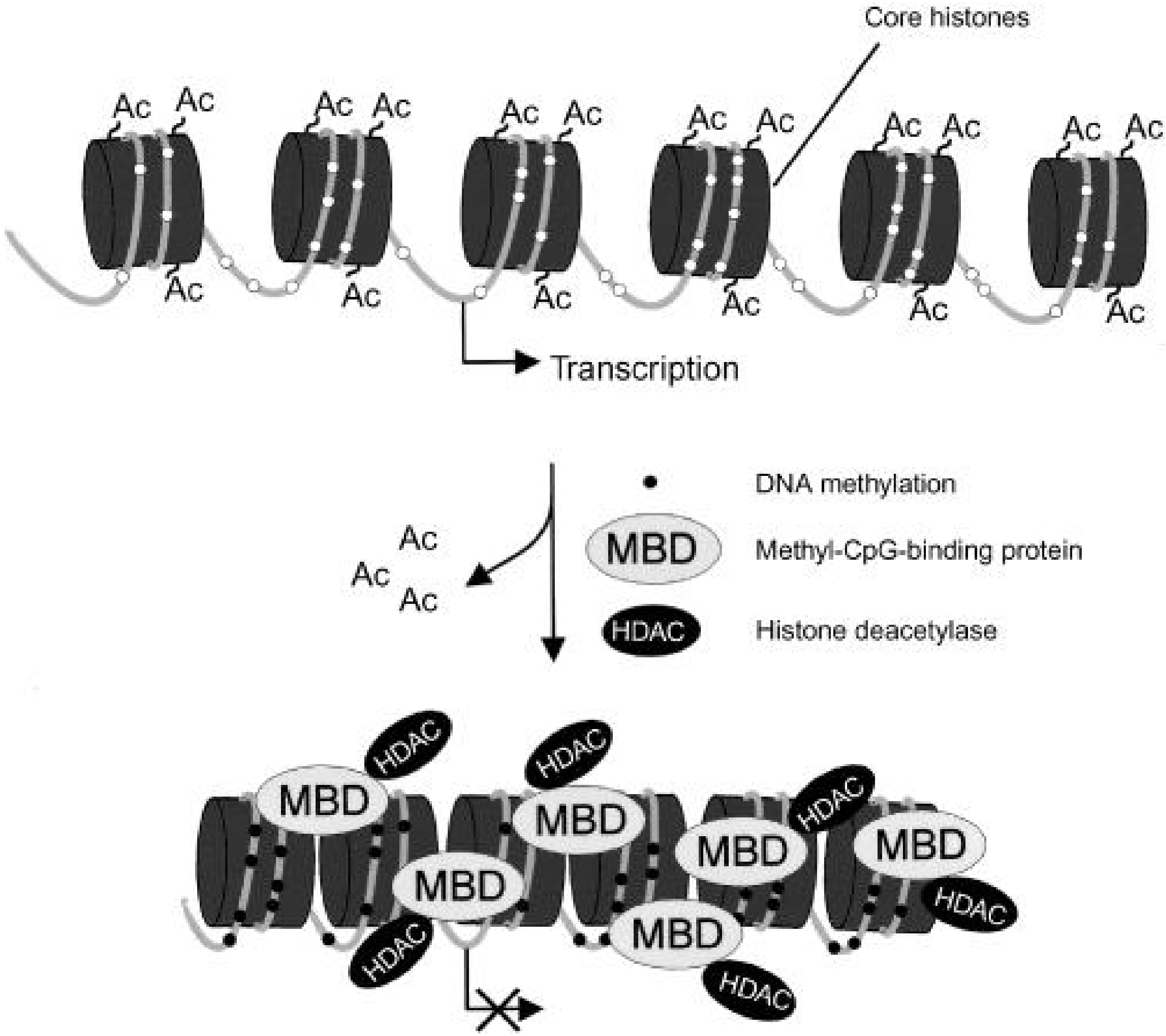
Methylation of CpG islands can block transcription by two distinct pathways:

- 1. Direct blocking of TFIID binding





2. Recruitment of histone deactylases



DNA methylation pattern can be inherited to daughter cells

```
   CH3
   |
CACTCGTCATT
GTGAGCAGTAA
       |
       CH3
```

Replication

```
   CH3
   |
CACTCGTCATT
GTGAGCAGTAA
```

```
CACTCGTCATT
GTGAGCAGTAA
       |
       CH3
```

DNA methyltransferase I

```
   CH3
   |
CACTCGTCATT
GTGAGCAGTAA
       |
       CH3
```

Histone modification pattern is also inherited to daughter cells through a poorly understood mechanism

- Since DNA methylation is at least sometimes linked to histone modification, conservation of histone pattern in daughter cells might be just a consequence of DNA methylation inheritance
- During replication, parental histones are randomly distributed to both daughter chromatides. Modified parental histones might serve as “nests” for modification of non-parental histones

Molecular mechanisms of transcription activation and repression

- 1. Chromatin mediated transcription control
- 2. Transcription control through the Mediator
- 3. Transcription control through DNA methylation
- 4. Regulation of transcription factor activity
- 5. Nuclear receptor

Regulation of transcription factor activity

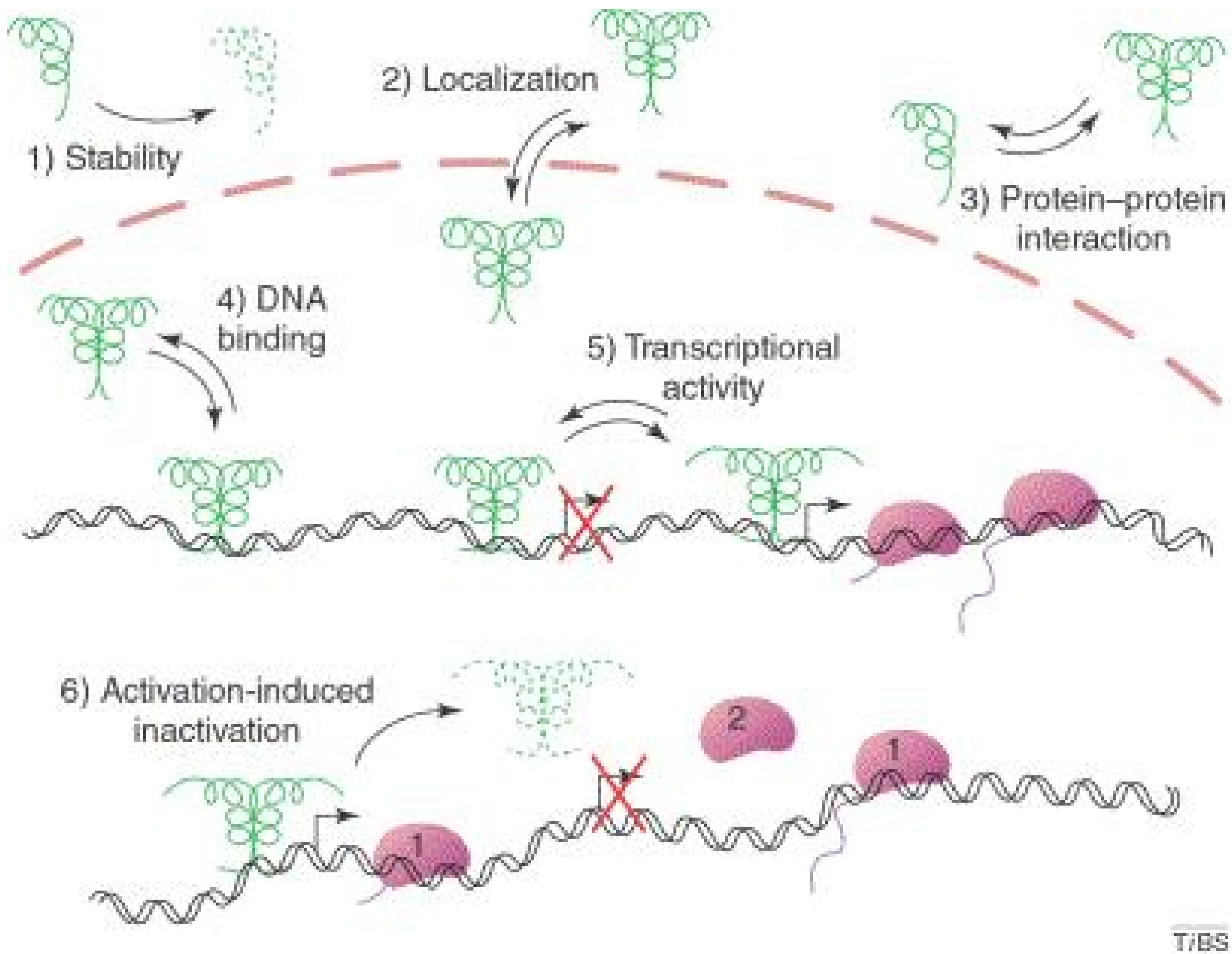
- The activity of transcription factors can be regulated by:

(1) covalent modification (<u>**phosphorylation**</u>, acetylation, ubiquitination, sumolation etc.)

(2) by binding to ligands (nuclear receptors)

Phosphorylation of transcription factors

Addition or removal of one or several phosphate groups on serine, threonine or tyrosine residues by a protein kinase or protein phosphatase.

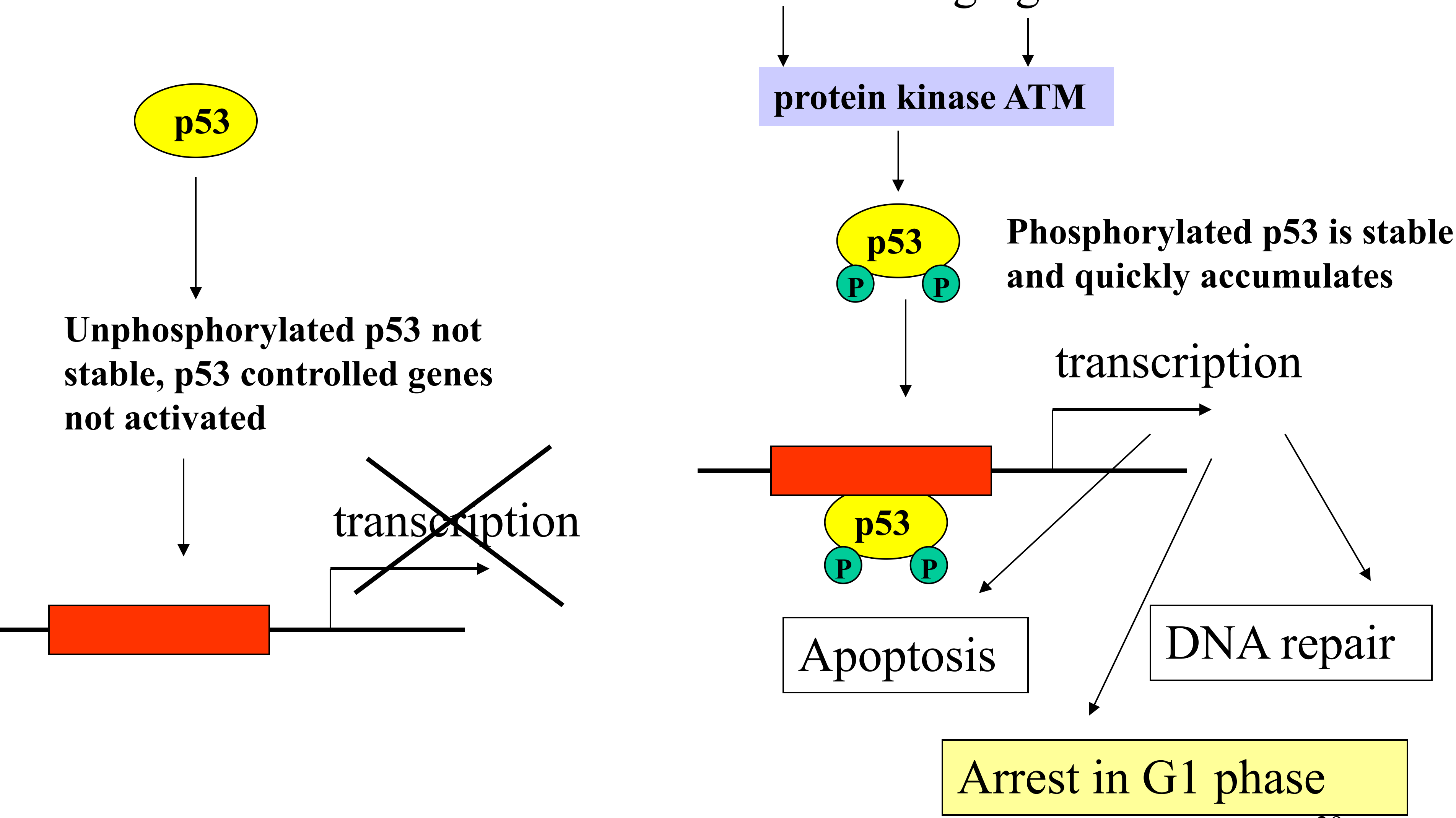


Protein p53

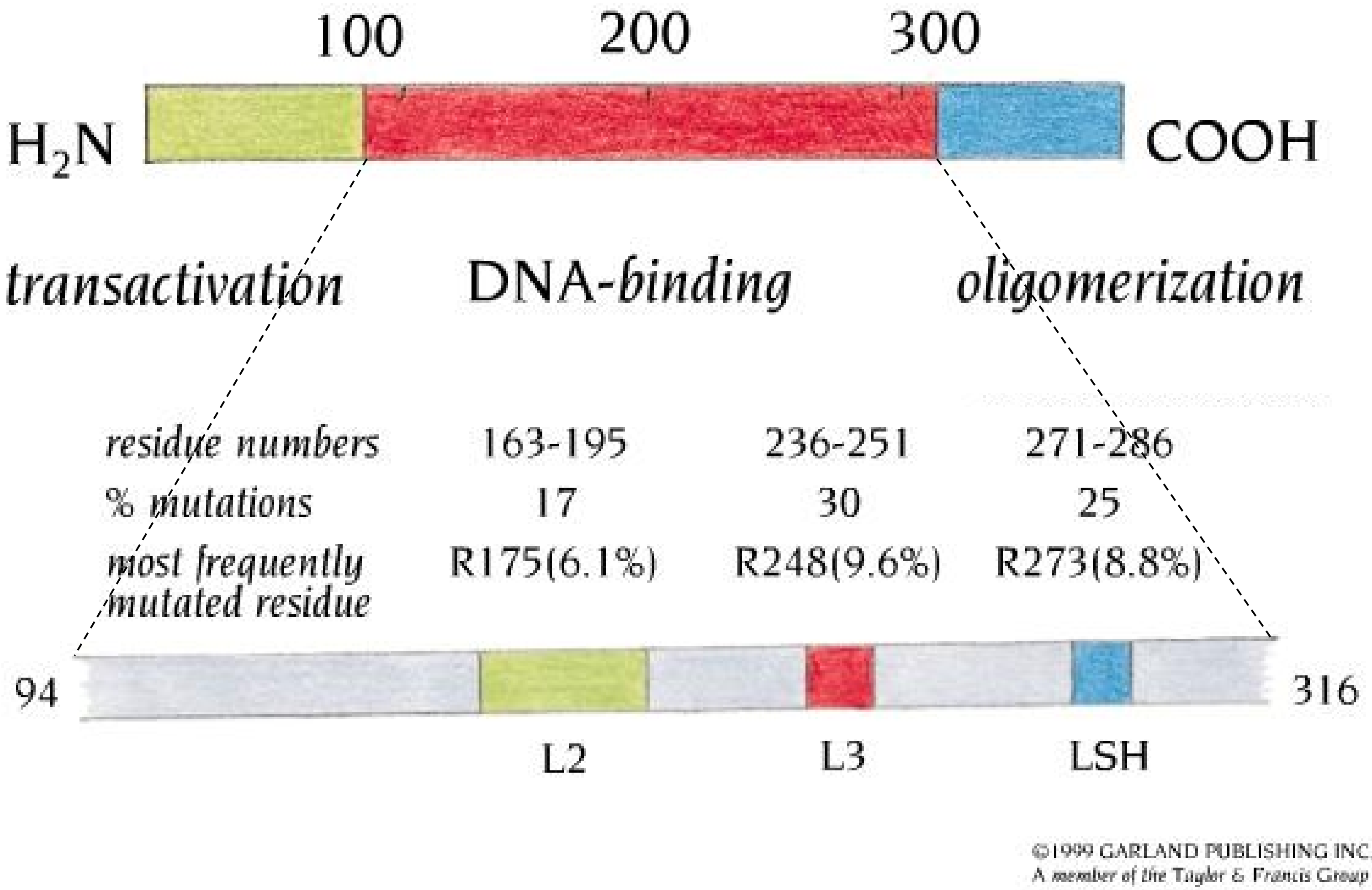
- p53 (**p**rotein **53** kDa) is one of the most frequently cited biomolecules in Science Citation index
- One of p53 functions is to act as a tumor supressor, which prevents cell division under DNA damaging conditions as exposure to UV light, etc
- Knockout mice lacking p53 show normal development, but show predisposition to develop multiple tumors
- About half of all 6.5 million people, annually diagnosed for various forms of cancer have mutations in p53 gene

Normal conditions

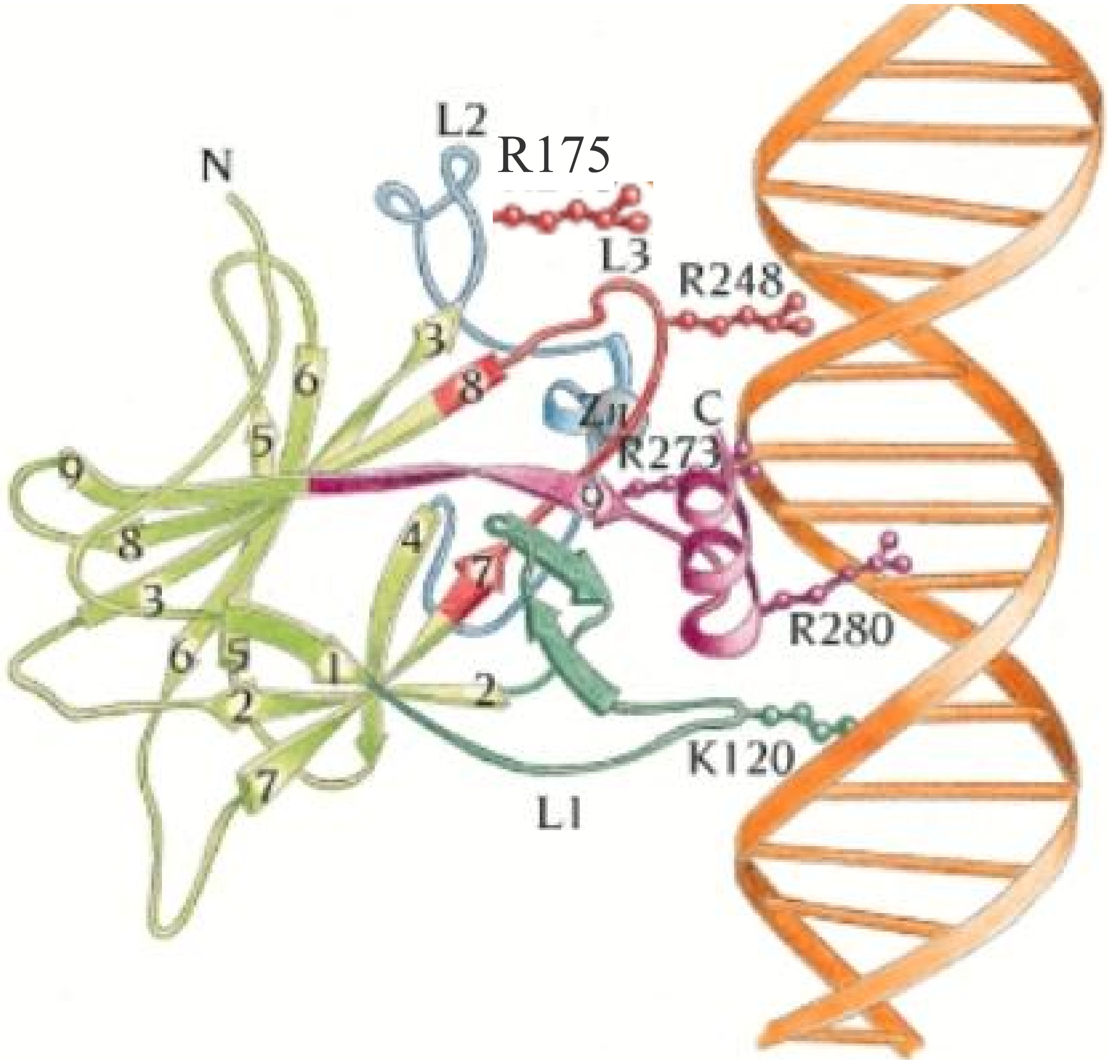
DNA-damaging conditions



Most p53 cancer associated mutations occur in DNA binding regions



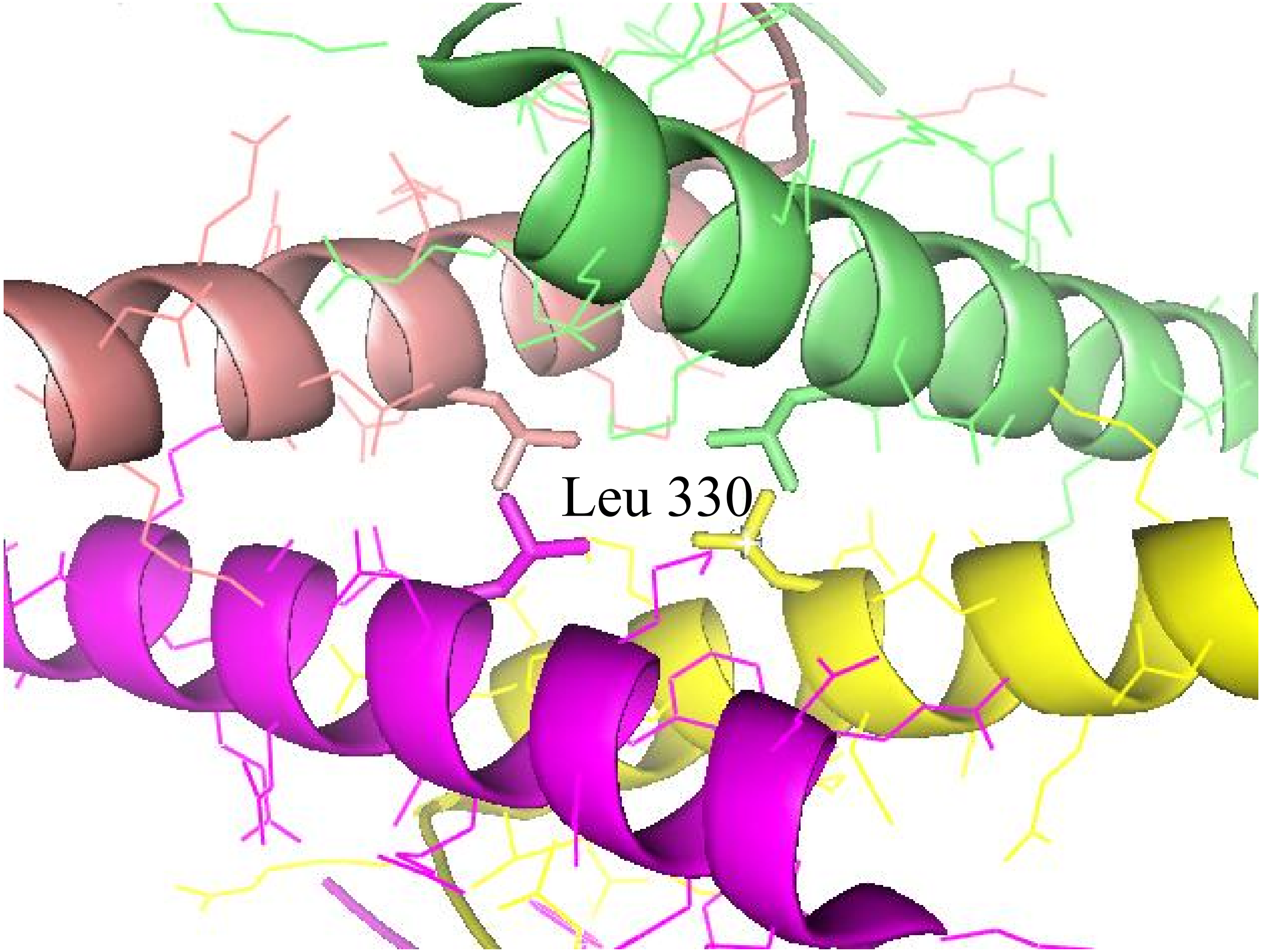
The 3D structure of p53 bound to DNA



K120, R280, R248, R273 –binds directly to DNA. Mutation of any of those leads to decreased affinity

R175 – holds the L3 loop in the correct conformation. Mutation of R175 leads to unfunctional L3 loop

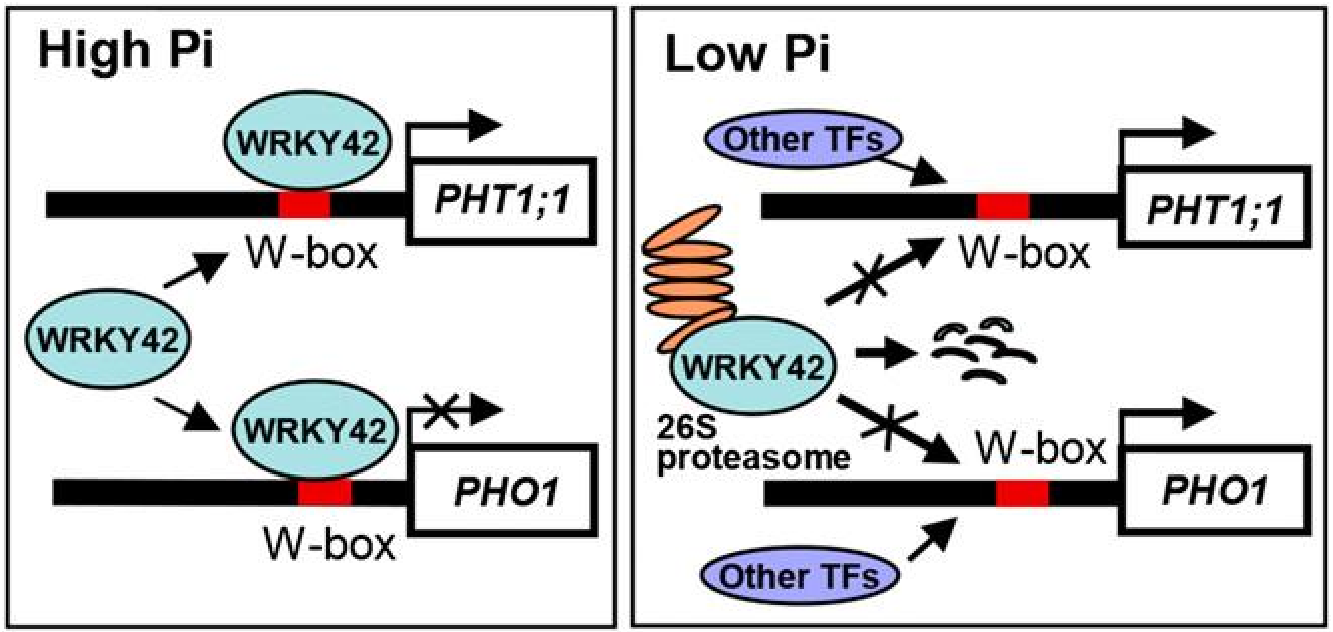
Some p53 cancer associated mutaions occur in tetramerization region



Ubiquitination of TFs

- TFs sometimes modified by a small protein called ubiquitin.
- Ubiquitination of some activators can have an activating effect, but polyubiquitination marks these same proteins for destruction.

AtWRKY42 modulated Pi homeostasis to adapt to environmental Pi availability



Chen et al. Plant Physiology, 2015

Sumoylation of TFs

- Sumoylation: the addition of one or more copies of the 101-aa polypeptide SUMO to lysine residues on a protein.
- Sumoylated activators appear to be targeted to a specific nuclear compartment that keeps them stable, but unable to reach their target genes.

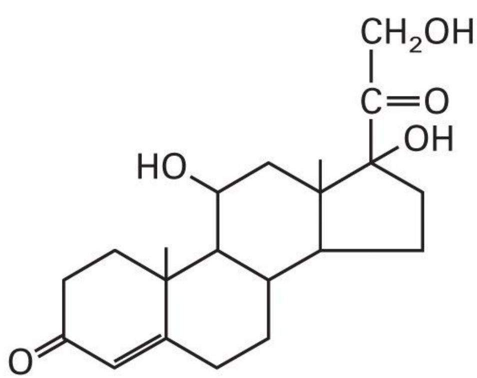
Acetylation of TFs

- We have known that histones can be acetylated on lysine residues by HATs, which decreases the histones' repressive activity.

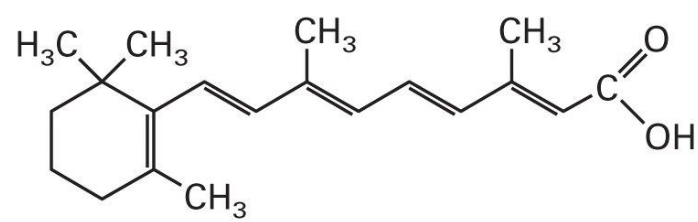
- Recently, it was reported that HATs can also acetylate nonhistone activators and repressor, and this can have either positive or negative effects on the acetylated protein's activities.

Nuclear receptors

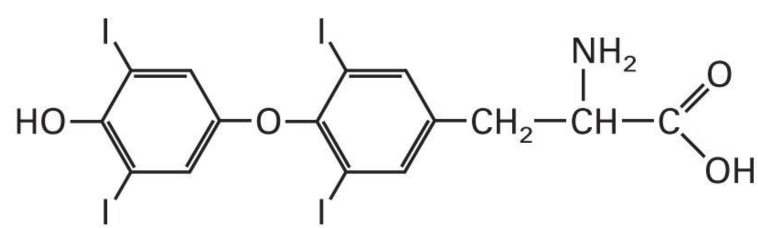
- Transcription factors which get activated by lipid-soluble hormones
- Lipid-soluble hormones – small hydrophobic molecules capable to diffuse freely through plasma and nuclear membranes



Cortisol



Retinoic acid

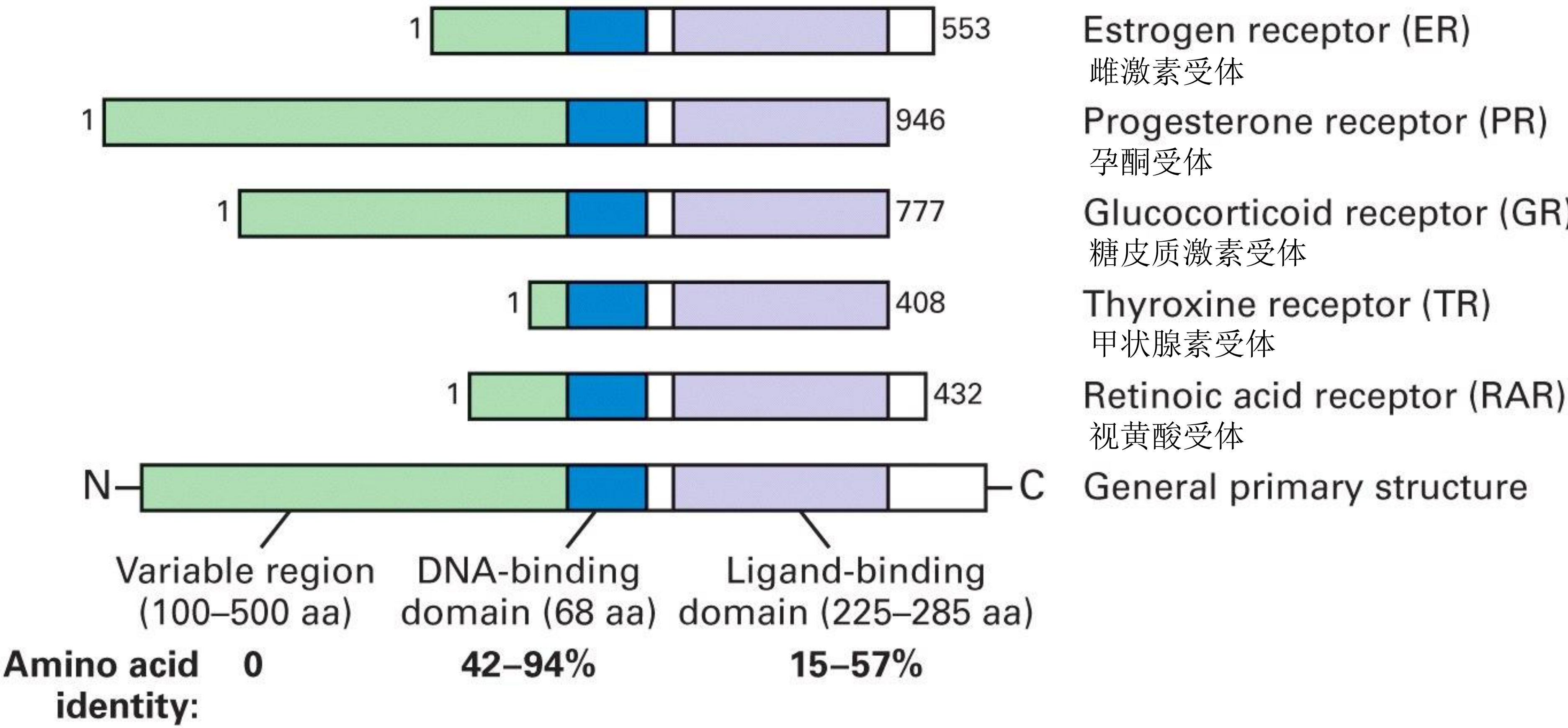


Thyroxine

Molecular mechanisms of transcription activation and repression

- 1. Chromatin mediated transcription control
- 2. Transcription control through the Mediator
- 3. Transcription control through DNA methylation
- 4. Regulation of transcription factor activity
- 5. Nuclear receptor

Domains of nuclear receptors



DNA binding domain – two C_4 zinc finger repeats

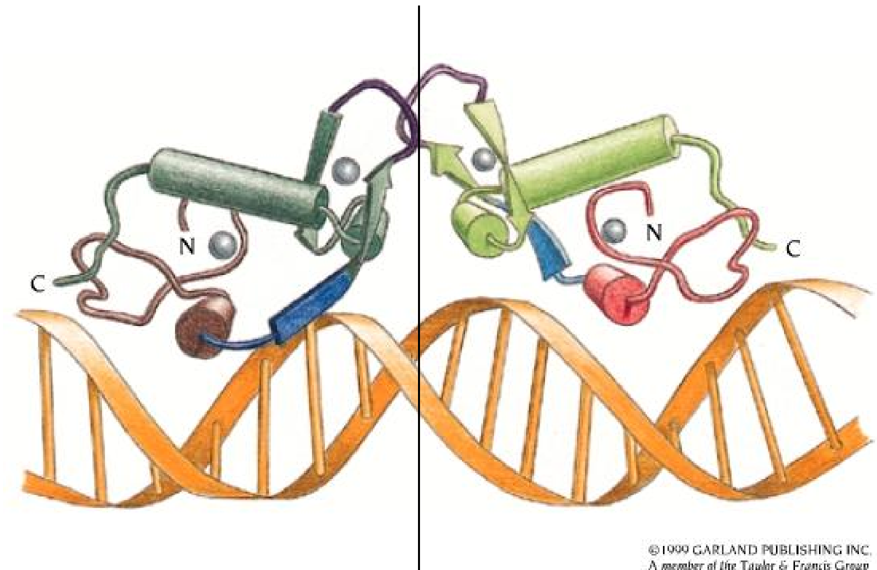
Homodimeric nuclear receptors are made of two identical subunits and bind to inverted DNA repeats

(a) **GRE**

5′ AGAACA$(N)_3$ TGTTCT 3′
3′ TCTTGT$(N)_3$ ACAAGA 5′

(b) **ERE**

5′ AGGTCA$(N)_3$ TGACCT 3′
3′ TCCAGT$(N)_3$ ACTGGA 5′



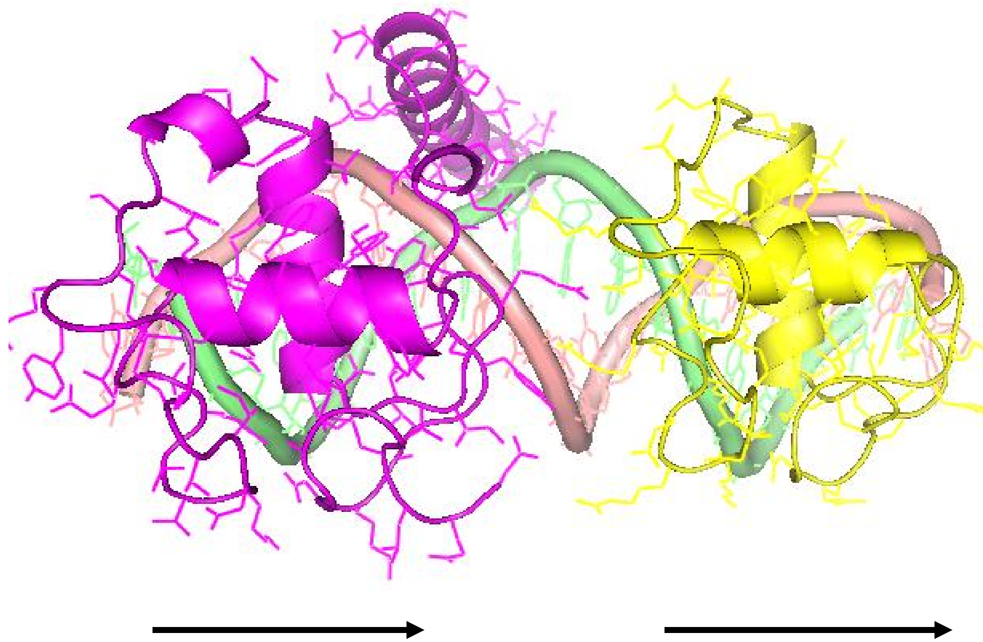
2-fold symmetry

Direct repeat binding sites of heterodimeric nuclear receptors

(c) **VDRE**
5′ AGGTCA$(N)_3$ AGGTCA 3′
3′ TCCAGT$(N)_3$ TCCAGT 5′

(d) **TRE**
5′ AGGTCA$(N)_4$ AGGTCA 3′
3′ TCCAGT$(N)_4$ TCCAGT 5′

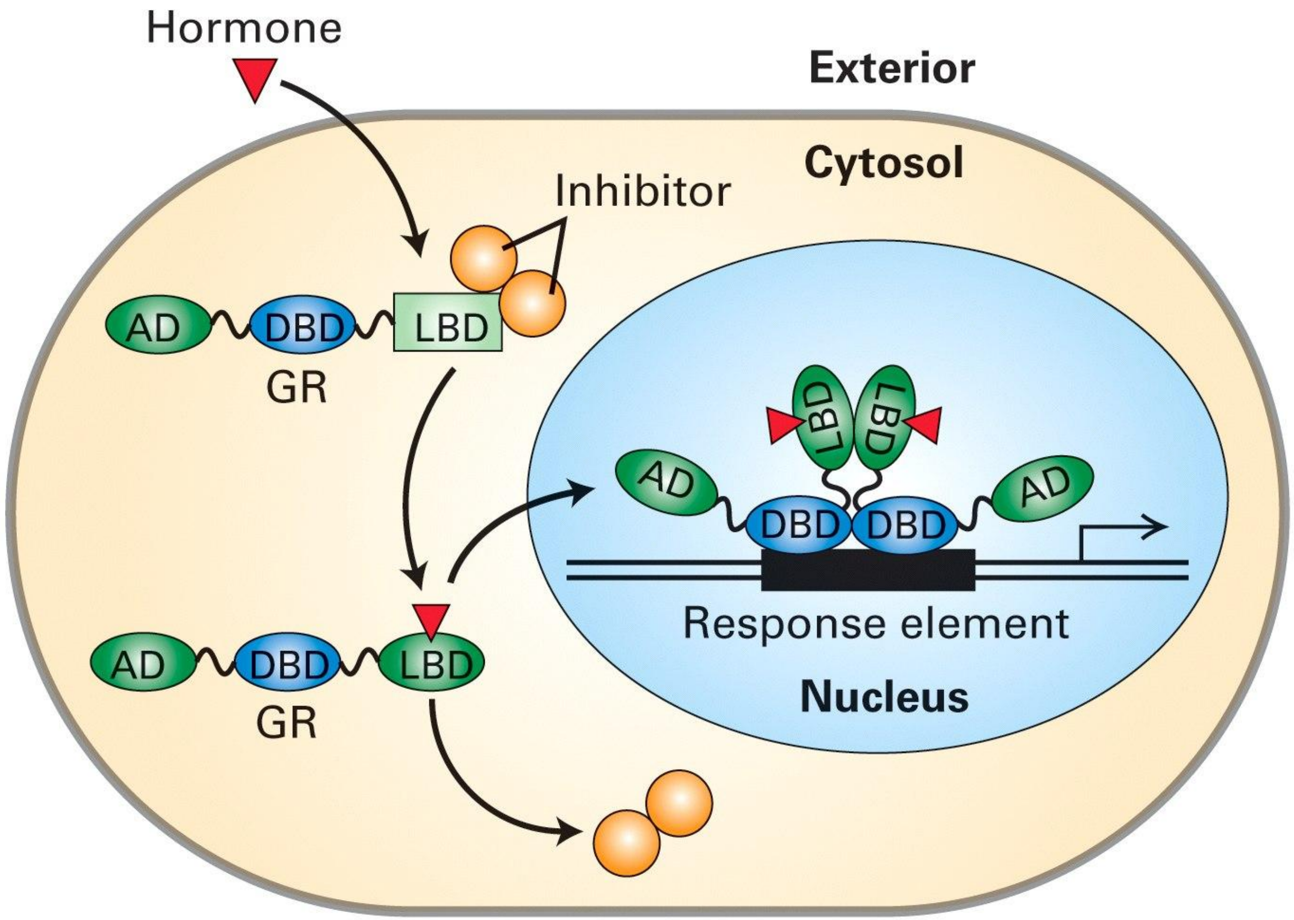
(e) **RARE**
5′ AGGTCA$(N)_5$ AGGTCA 3′
3′ TCCAGT$(N)_5$ TCCAGT 5′



Heterodimeric nuclear receptors are made of two different subunits, one of them always being an universal monomer, called RXR

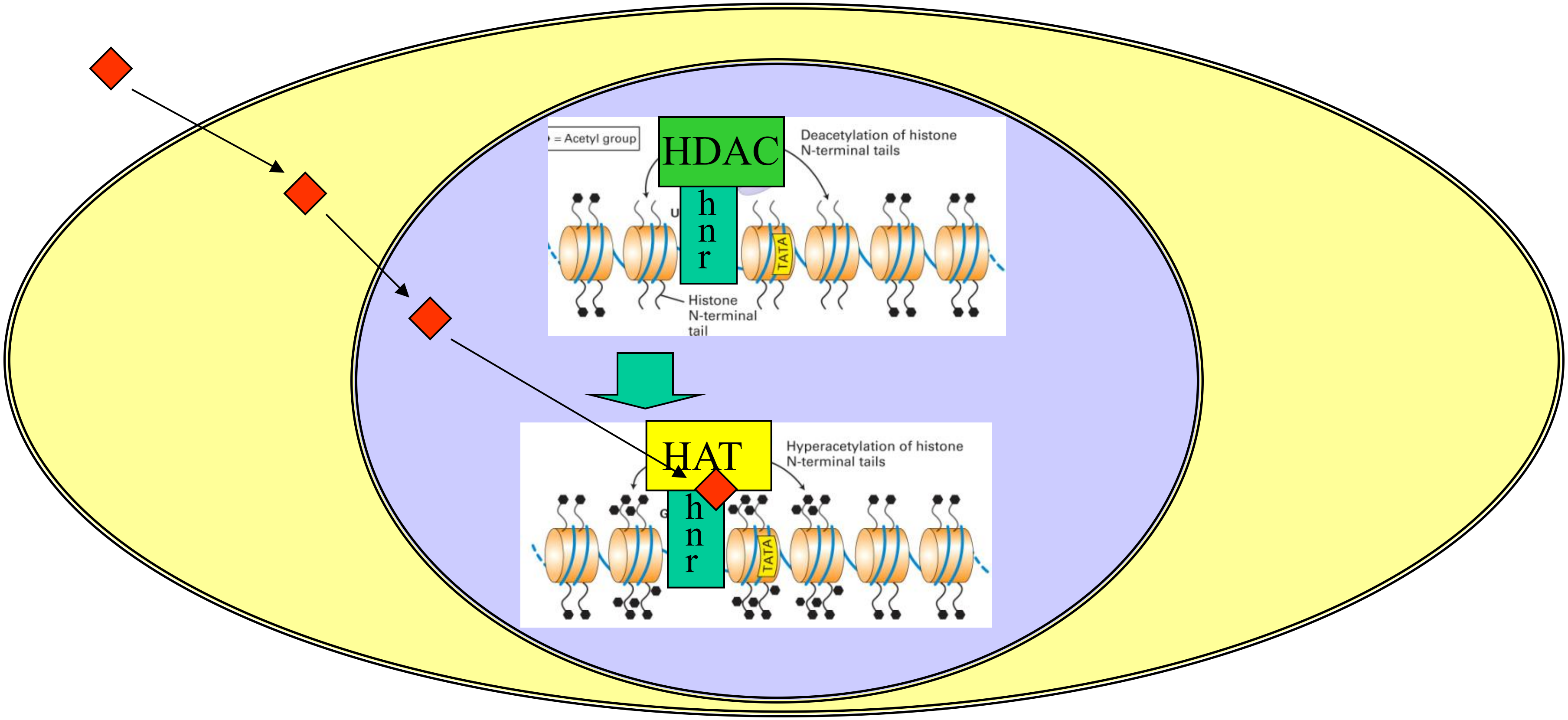
Binding specificity of heterodimeric nuclear receptors is achieved solely by variable length of spacer nucleotide sequence

Action of homodimeric nuclear receptors



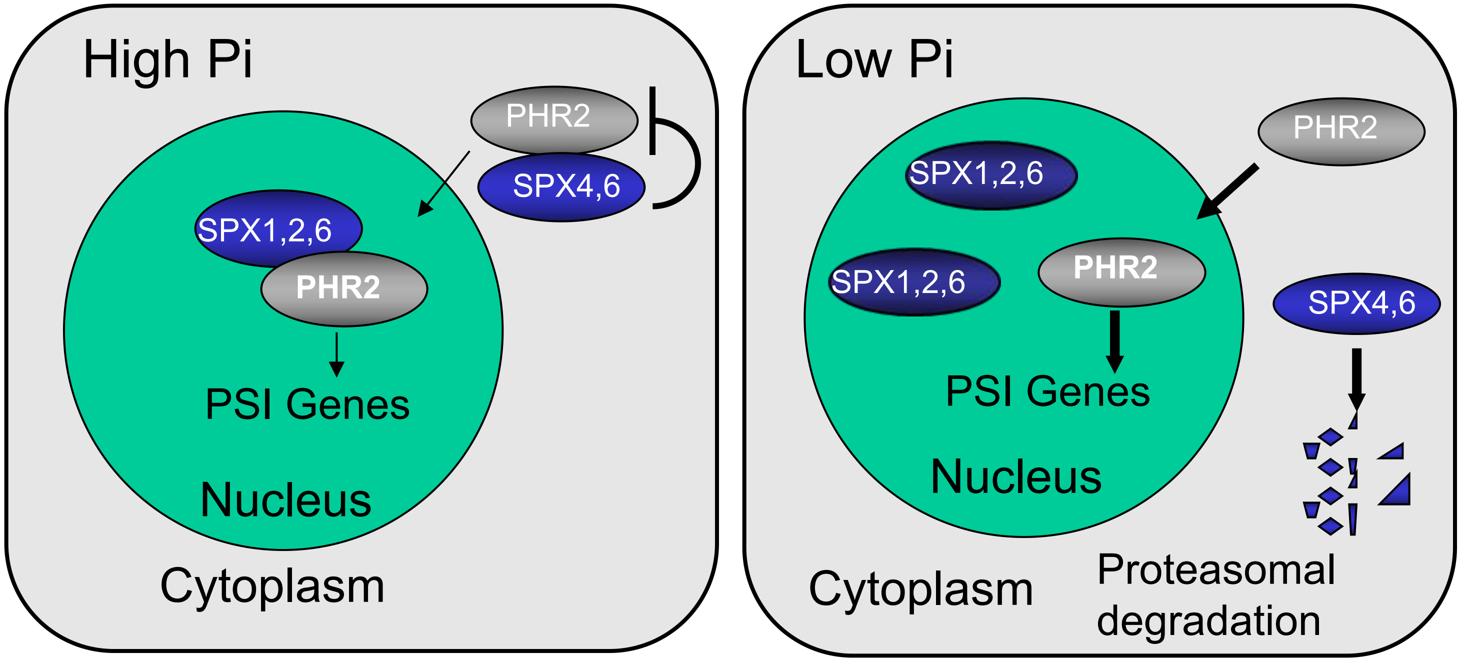
- In the absence of hormone, nuclear receptor is located in cytoplasm
- Upon binding to hormone, the nuclear receptor gets transported to nucleus, where it binds to the response element

Action of **h**eterodimeric **n**uclear **r**eceptors



- In the absence of hormone, hnr binds to DNA response element and recruits histone deacetylases. Transcription is blocked.
- When hormone diffuses into the nucleus and binds to hnr, histone deacetylase gets released and histone acetylase binds instead. Transcription is activated.

Regulating the function of transcription factor by an interaction protein



Wang et al. 2014 PNAS
Lv et al. 2014 Plant Cell
Zhong et al. 2018 New Phytol

Methods for detecting changes in transcription level

- Northern blot
- RT-PCR
- Real time quantitative RT-PCR
- *In situ* hybridization (ISH, FISH)
- Micro-array
- Deep sequencing (transcriptome)

Methods for detecting DNA protein interaction

- Chromatin immunoprecipitation (ChIP)
- Gel shift assay (Electrophoretic mobility shift assay, EMSA)
- DNase footprinting
- Filter binding
- Y1H

  …

Chromatin Immunoprecipitation(ChIP)

Chromatin immunoprecipitation, or ChIP, refers to a procedure used to determine whether a given protein binds to or is localized to a specific DNA sequence *in vivo*.

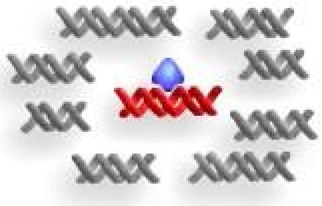
Applications of ChIP, ChIP-Chip, and ChIP-Seq

- **Target Gene Identification (individual and global)**
- **Binding Site Consensus Identification**
- **DNA (RNA)--Protein Interactions**
- **Analysis of Epigenetic Phenomenon (eg, methylation, silencing)**

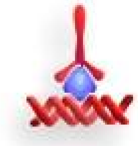
Procedure

1 

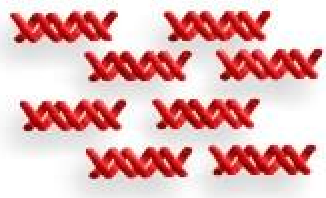
DNA-binding proteins are crosslinked to DNA with formaldehyde in vivo.

2 

Isolate the chromatin. Shear DNA along with bound proteins into small fragments. (Sonication)

3 

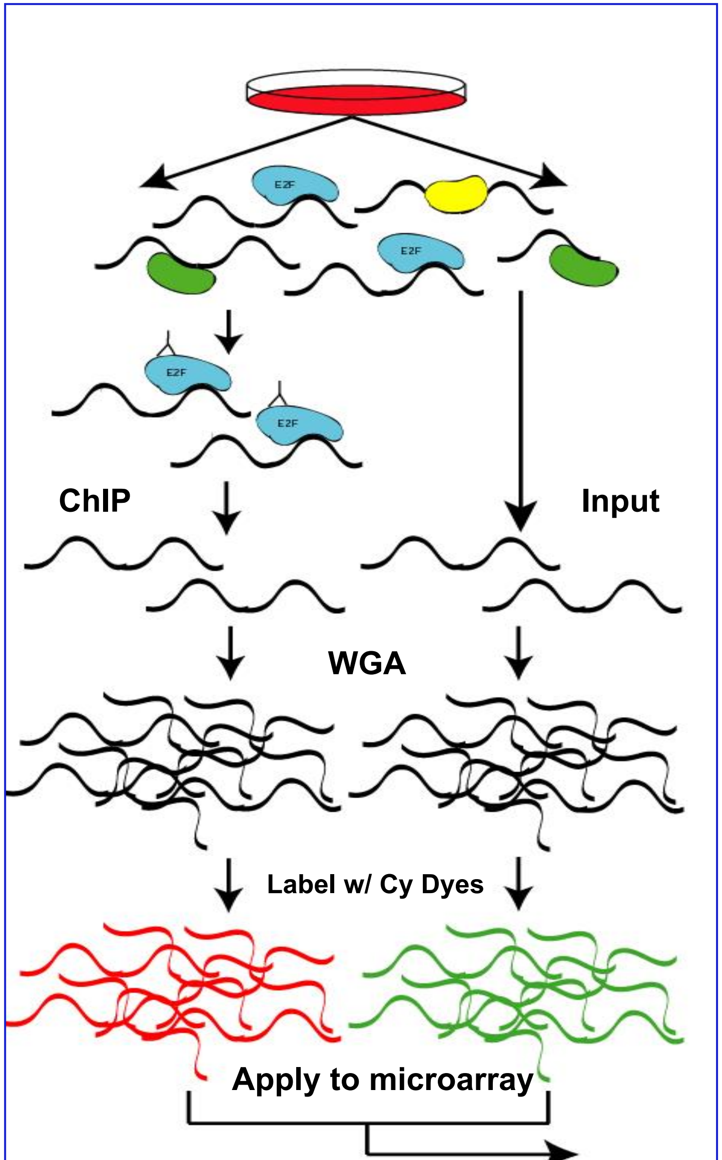
Bind antibodies specific to the DNA-binding protein to isolate the complex by precipitation. Reverse the cross-linking to release the DNA and digest the proteins.

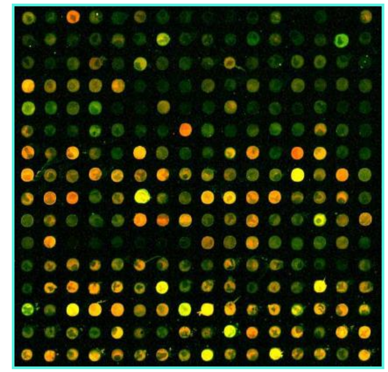
4 

Use PCR to amplify specific DNA sequences to see if they were precipitated with the antibody.

(PCR; ChIP on chip; Sequencing)

ChIP-chip





Compare Red/Green signal intensities to identify binding sites

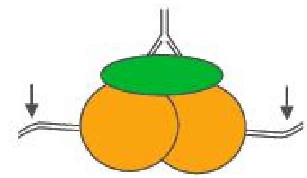
Peak Finding a challenge, but commercial software and local tools becoming more available

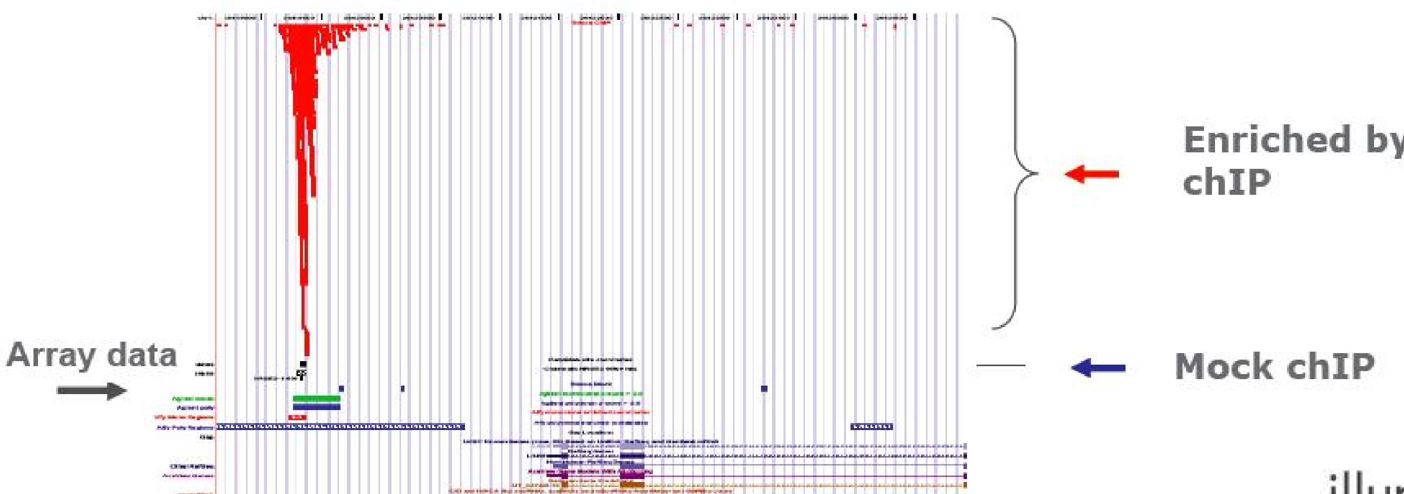
ChIP-Seq

Genomic tag sequencing

Discovering transcription factor binding sites using chIP

- Treat chromatin with antibody specific to TF and recover trapped DNA
- Generate 6M tags by Solexa sequencing
- Identify clustered alignments in human genome
- Compare to microarray data in ENCODE region
- Genome-wide dataset





17

ENCODE: Encyclopedia of DNA Elements

Student's presentation (today)

Paper:

DNA replication–coupled histone modification maintains Polycomb gene silencing in plants. Jiang et al., Science 357, 1146–1149 (2017)